

# Bayan: benchmark report

Every simplifier we built, on BayanBench v2.0, with the scorer checked against human raters

Bayan team (Cogni-Libya) · report generated from docs/benchmark\_report

4 October 2026

## Abstract

We compare 35 systems: copying the input, the two models in today's app, models 1 and 2, the model 3 family (AraT5v2 trained on corpus v1) with every decoding, precision and safety-net variant, a fluency retrain of model 3, and two dropped attempts (student DPO and minimum-risk training). All are scored with the frozen BayanBench v2.0 rules on the dev and test splits, with meaning judged by Gemma 4 31B (AUC 0.85 against the human majority). The candidate to ship, **model 3, app int8, beam 4 + safety net**, is *at least as faithful* as today's app on test (P(same) 0.83 vs 0.79; paired difference +0.04 [-0.00, +0.08], an interval that just touches zero), keeps every number (100% vs 84%), changes no negation, limit or condition, and leaves easy and protected text alone far more often. It shortens the longest clause clearly more (5.4 vs 1.9 words; paired +3.5 [+2.5, +4.5]), but its reading-level drop is not significantly larger on test (0.20 vs 0.14). Two caveats run through the report: int8 quantization costs about half of model 3's float simplification, and the ship candidate reaches its meaning score partly by leaving more text unchanged (47% vs 19%); on the outputs it does change, P(same) is 0.67 vs 0.74 for the app.

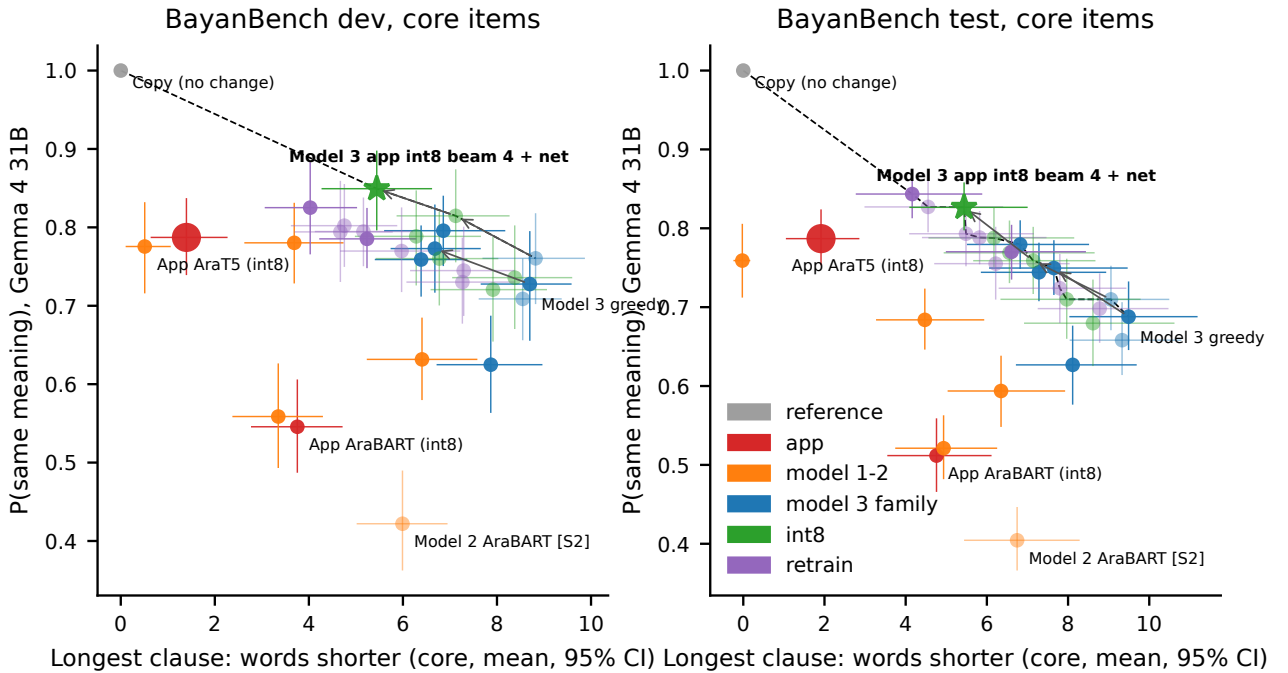
## Contents

|    |                                              |    |
|----|----------------------------------------------|----|
| 1  | Summary                                      | 3  |
| 2  | Systems                                      | 4  |
| 3  | Benchmark, scorer and statistics             | 5  |
| 4  | Can we trust the scorer?                     | 5  |
| 5  | Results: leaderboards                        | 7  |
| 6  | Meaning                                      | 11 |
| 7  | Simplicity                                   | 14 |
| 8  | Safety and restraint                         | 14 |
| 9  | Arabic quality (diagnostic)                  | 15 |
| 10 | Precision, decoding and the safety net       | 16 |
| 11 | Item sets                                    | 17 |
| 12 | Reference-based scores                       | 17 |
| 13 | Human ratings                                | 18 |
| 14 | Training                                     | 18 |
| 15 | On-device cost                               | 19 |
| 16 | Ship gate: the candidate against today's app | 20 |

|                                                               |           |
|---------------------------------------------------------------|-----------|
| <b>17 Examples</b>                                            | <b>21</b> |
| <b>18 Limitations</b>                                         | <b>23</b> |
| <b>A Per-set leaderboards</b>                                 | <b>24</b> |
| <b>B Paired differences against today's app, every system</b> | <b>29</b> |
| <b>C Reproducing this report</b>                              | <b>30</b> |

# 1 Summary

- **Recommendation: ship model 3, app int8, beam 4 + safety net** (Congi-libya/bayan-arat5-v1-bundle, 471 MB): the best balance we have. Against today’s app it is never significantly worse on meaning, significantly better on every safety measure, and significantly simpler on clause length on both splits (Table 11, Figure 24).
- **Test, core items, paired against today’s app:**  $P(\text{same}) +0.04 [-0.00, +0.08]$ ; meaning kept +4 [+0, +8] points; clause +3.5 [+2.5, +4.5] words; reading level +0.07 [-0.01, +0.15] levels; numbers kept 100% vs 84%; easy text left unchanged 74% vs 48%.
- **Dev agrees and is a little stronger:**  $P(\text{same}) +0.06 [-0.00, +0.13]$ ; clause +4.0 [+2.6, +5.4] words; level +0.15 [+0.04, +0.26] (significant on dev).
- **The cost of the phone:** float model 3 with the net shortens the longest clause by 7.7 words and drops 0.38 levels; the int8 bundle with beam 4 and the net, 5.4 and 0.20. Beam 4 and the net buy back meaning, not simplicity (Figure 16).
- **Copying inflates  $P(\text{same})$ :** a copy counts as perfect meaning. The ship candidate leaves 47% of core test items unchanged (app: 19%). On the outputs it does change, its  $P(\text{same})$  is 0.67 vs 0.74 for the app’s (Figure 9): when it rewrites, it rewrites harder and loses more meaning per rewrite. The net and the scorer catch the dangerous cases (numbers, negations), not every omission.
- **The fluency retrain is not worth shipping:** small gains in Gemma’s Arabic score, more copying, and under int8 it simplifies less than model 3.
- **Dropped:** student DPO (better proxies, worse on the 31B scorer) and MRT (the judge in the loss made outputs simpler but not more faithful).
- **int8 needs a VNNI-class CPU:** the app’s per-channel int8 recipe gives nonsense on an AVX2-only Xeon and fluent output on an AVX-VNNI CPU; ARM phones use the same integer dot-product path as VNNI (Table 5), but this still has to be confirmed on a phone.



**Figure 1:** Meaning against simplicity for every system (core items; 95% cluster-bootstrap intervals). Dashed: the Pareto frontier (no system is both more faithful and simpler). Star: the ship candidate. Arrows: adding the safety net to float model 3; float beam 4 → app int8 beam 4 → + net.

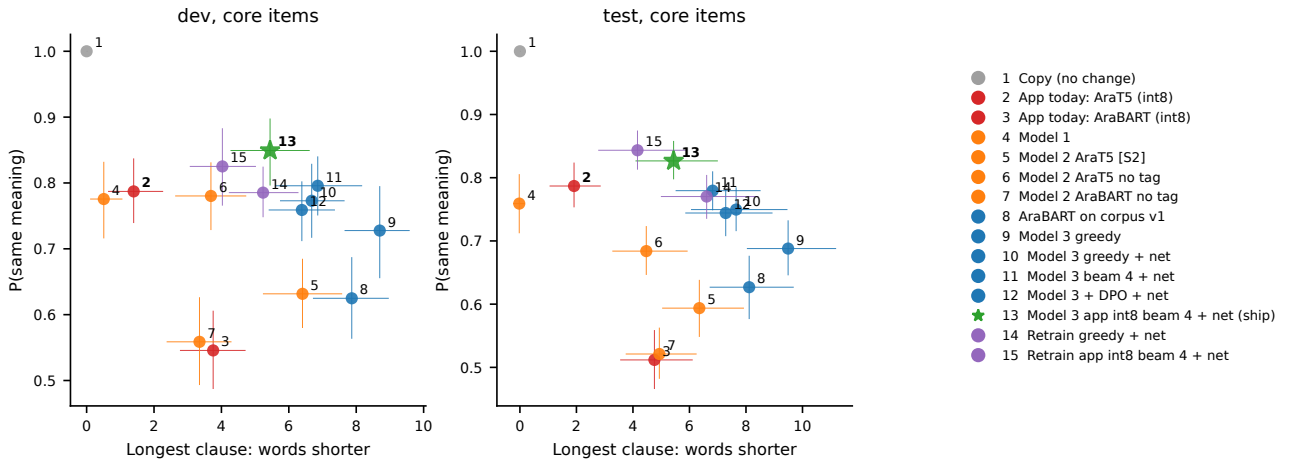


Figure 2: The same, main systems only, numbered (95% intervals).

## 2 Systems

All model 3-family systems use the prefix « بسط : » (no shadda); model 2 AraT5 used « بسط : ». “App int8” is the app’s own export recipe (optimum export, merged decoder, dynamic QInt8 per-channel on MatMul and Gather), run with the app’s pipeline (sentence pieces, the app’s guards). “Net” is the per-sentence safety net: a sentence that drops a number or a Latin token, or changes the count of negation, limit or condition words, is shown unchanged. “Hybrid  $p\%$ ” runs beam 4 only on the  $p\%$  least confident sentences.

| System                               | Group          | What it is                                    |
|--------------------------------------|----------------|-----------------------------------------------|
| Copy (no change)                     | reference      | the selection unchanged                       |
| App today: AraT5 (int8)              | app            | shipped app, model 2 AraT5v2, int8, no tag    |
| App today: AraBART (int8)            | app            | shipped app, model 2 AraBART, int8, no tag    |
| Model 1                              | model 1-2      | AraT5v2 on SAMER                              |
| Model 2 AraT5 [S2]                   | model 1-2      | model 2 AraT5v2, strength tag [S2], float     |
| Model 2 AraT5 no tag                 | model 1-2      | model 2 AraT5v2, no tag, float                |
| Model 2 AraBART [S2]                 | model 1-2      | model 2 AraBART, tag [S2], float              |
| Model 2 AraBART no tag               | model 1-2      | model 2 AraBART, no tag, float                |
| AraBART on corpus v1                 | model 3 family | AraBART, corpus v1, MLE (MRT baseline)        |
| Model 3 greedy                       | model 3 family | AraT5v2 on corpus v1, float, greedy           |
| Model 3 greedy + net                 | model 3 family | float, greedy, safety net                     |
| Model 3 beam 4                       | model 3 family | float, beam 4                                 |
| Model 3 beam 4 + net                 | model 3 family | float, beam 4, safety net                     |
| Model 3 + DPO                        | model 3 family | model 3 + student DPO (dropped)               |
| Model 3 + DPO + net                  | model 3 family | model 3 + DPO, safety net (dropped)           |
| Model 3 int8 per-tensor              | int8           | per-tensor int8, greedy                       |
| Model 3 int8 per-tensor + net        | int8           | per-tensor int8, greedy, net                  |
| Model 3 app int8 greedy              | int8           | app recipe (per-channel int8), greedy         |
| Model 3 app int8 greedy + net        | int8           | app int8, greedy, net                         |
| Model 3 app int8 beam 4              | int8           | app int8, beam 4                              |
| Model 3 app int8 beam 4 + net (ship) | int8           | app int8, beam 4, net: the shipping candidate |
| Retrain greedy                       | retrain        | fluency retrain, float, greedy                |

| System                        | Group   | What it is                                                 |
|-------------------------------|---------|------------------------------------------------------------|
| Retrain greedy + net          | retrain | float, greedy, net                                         |
| Retrain beam 4                | retrain | float, beam 4                                              |
| Retrain beam 4 + net          | retrain | float, beam 4, net                                         |
| Retrain app int8 greedy       | retrain | app int8, greedy                                           |
| Retrain app int8 greedy + net | retrain | app int8, greedy, net                                      |
| Retrain app int8 beam 4       | retrain | app int8, beam 4                                           |
| Retrain app int8 beam 4 + net | retrain | app int8, beam 4, net                                      |
| Model 3 hybrid 0% beam        | hybrid  | float greedy, beam 4 on the 0% least confident sentences   |
| Model 3 hybrid 10% beam       | hybrid  | float greedy, beam 4 on the 10% least confident sentences  |
| Model 3 hybrid 20% beam       | hybrid  | float greedy, beam 4 on the 20% least confident sentences  |
| Model 3 hybrid 30% beam       | hybrid  | float greedy, beam 4 on the 30% least confident sentences  |
| Model 3 hybrid 50% beam       | hybrid  | float greedy, beam 4 on the 50% least confident sentences  |
| Model 3 hybrid 100% beam      | hybrid  | float greedy, beam 4 on the 100% least confident sentences |

### 3 Benchmark, scorer and statistics

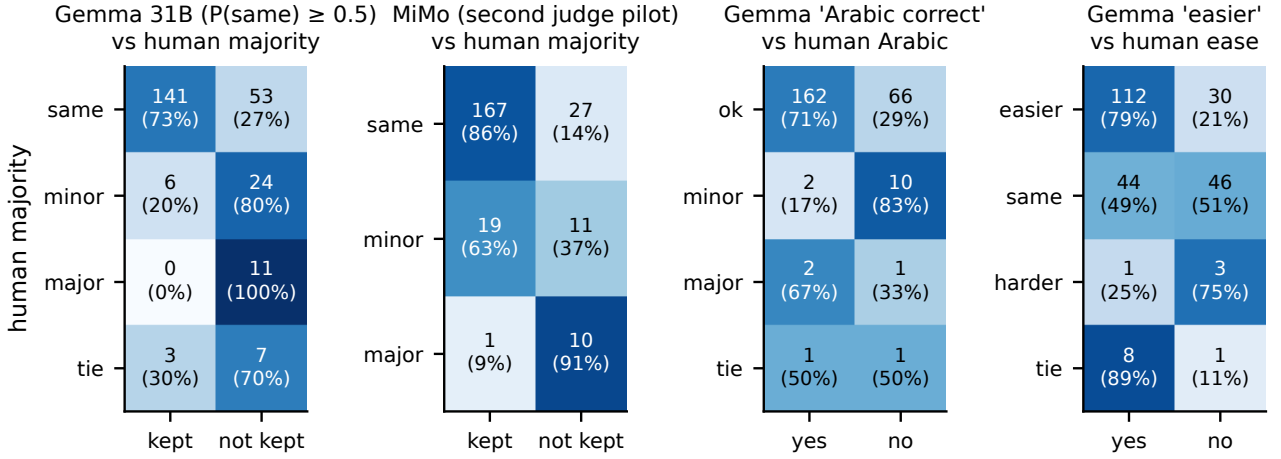
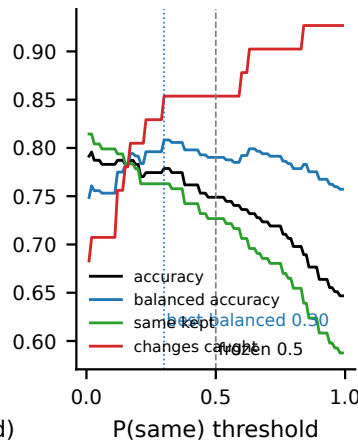
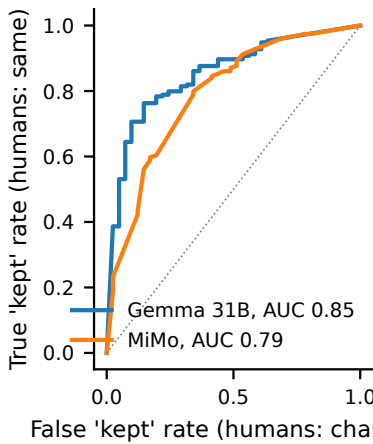
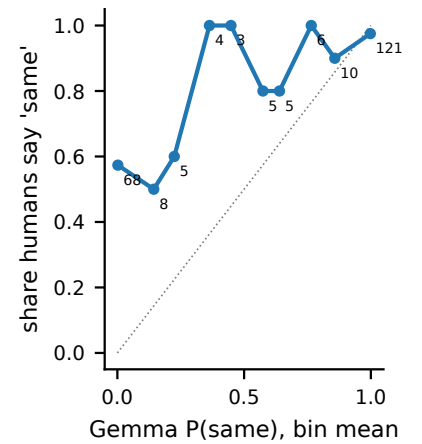
- **BayanBench v2.0** (frozen 4 Oct; data v1.0, 731 dev and 1,250 test items). Item sets: *core* (BAREC sentences), *held-out* (news and legal), *outside* (real text from outside BAREC and SAMER) and *written* (medicine leaflets). Tracks: A1–A3 simplify, B hard text, C easy text (should stay), E1–E4 edge cases (cut-off endings, lists, dialogue), F protected text (scripture).
- **Meaning** is Gemma 4 31B (int4) in scoring mode: P(yes) for *same*, *added*, *missing* and *contradict*. The official rule: *meaning kept* =  $P(\text{same}) \geq 0.5$  and every number kept.
- **Simpler** is measured as change: words cut from the longest clause (items with a clause over 15 words), BAREC reading levels dropped by the hardest sentence (items at level 10+), and SAMER level-4/5 words removed.
- **Joint** (meaning kept and simpler) uses our own minimum change, 2 clause words or half a level; v2.0 leaves it undecided.
- **Extra questions** (diagnostic): Gemma 31B’s P(yes) for “Arabic correct”, “easier to read” and “coherent”. Their agreement with raters is weak (Section 4), so they are never used for decisions.
- **Intervals** are 95% cluster-bootstrap intervals over documents (bayanbench’s own `stats.ci`); paired differences use the same items for both systems. In tables,  $\uparrow/\downarrow$  mark paired differences whose interval excludes zero.

### 4 Can we trust the scorer?

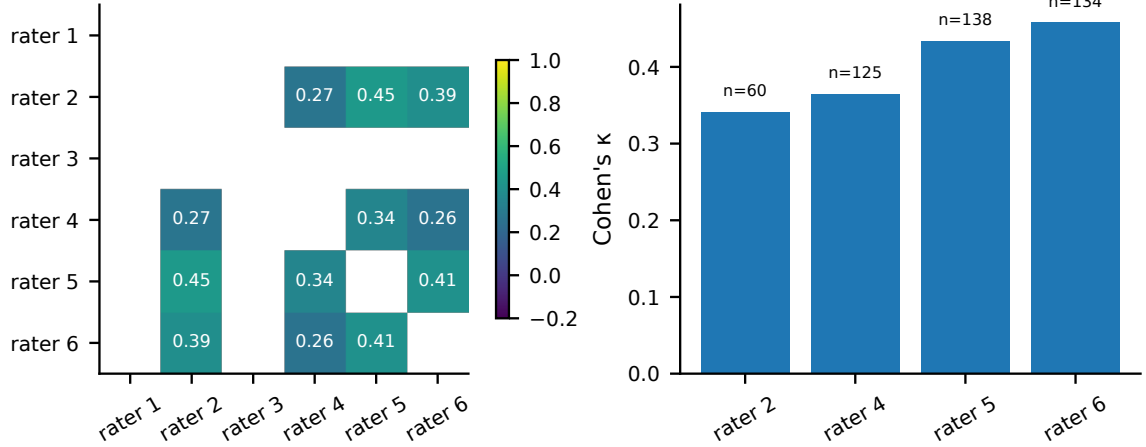
Six raters rated 290 tasks in the BayanBench v2 round (3 Oct): 245 single outputs (meaning, Arabic, ease) and 45 side-by-side pairs. Table 2 and Figures 3–5 compare the scorer with the human majority.

**Table 2:** The scorer against human raters.

| Comparison                                   | Statistic                                          | Value              | Note                                           |
|----------------------------------------------|----------------------------------------------------|--------------------|------------------------------------------------|
| Gemma 31B P(same) vs majority “same”         | AUC                                                | 0.852              | 194 same / 41 changed                          |
| Gemma 31B P(same) vs majority “major”        | AUC                                                | 0.922              |                                                |
| At the frozen threshold 0.5                  | accuracy / balanced                                | 0.749 / 0.790      | same kept 73%, changes caught 85%              |
| Best balanced threshold                      | 0.30                                               | 0.808              |                                                |
| Gemma “Arabic correct” vs human Arabic ok    | AUC                                                | 0.769              | n=243                                          |
| Gemma “easier” vs human easier               | AUC                                                | 0.697              | n=236                                          |
| MiMo (second-judge pilot) vs majority “same” | AUC                                                | 0.790              | sided with humans on 31/62 Gemma disagreements |
| Inter-rater agreement, meaning               | Krippendorff $\alpha$ (nominal / ordinal / binary) | 0.30 / 0.38 / 0.32 | 87 items with 2+ raters                        |
| Inter-rater agreement, Arabic                | Krippendorff $\alpha$ (nominal / ordinal / binary) | 0.15 / 0.17 / 0.17 | 87 items with 2+ raters                        |
| Inter-rater agreement, ease                  | Krippendorff $\alpha$ (nominal / ordinal / binary) | 0.26 / 0.22 / 0.26 | 87 items with 2+ raters                        |

**Figure 3:** Confusion matrices against the human majority (rows; “tie” = no majority). Counts and row percentages.**ROC: meaning kept vs human majority ‘same’ Threshold sweep****Calibration (n per bin)****Figure 4:** Left: ROC of P(same) (and of the MiMo second-judge pilot) for the human majority “same”. Middle: how accuracy, balanced accuracy, same-kept rate and changes-caught rate move with the threshold. Right: calibration.

Rater pairs: Cohen's  $\kappa$  on 'same meaning' ( $\geq 15$  shared items) Each rater vs Gemma (kept at 0.5)



**Figure 5:** Agreement between raters, and of each rater with the scorer (binary: same vs changed).

**Reading.** The meaning scorer separates “same” from “changed” well (AUC 0.85), and the frozen 0.5 threshold gives a balanced accuracy close to the best cut’s (0.79 at 0.5 vs 0.81 at 0.30; Table 2): at 0.5 the scorer lets about a quarter of the human-“same” pairs fail, so it errs towards strictness. Humans agree only moderately with one another on meaning ( $\alpha = 0.32$  binary), and barely on Arabic ( $\alpha = 0.17$ ) and ease ( $\alpha = 0.26$ ): this is why the extra questions stay diagnostic. MiMo failed the second-judge rule (it sided with the humans on only half of Gemma’s disagreements).

## 5 Results: leaderboards

**Bold:** best in the column (copy excluded). Kept, Joint, Num., Easy=, Prot. and Neg. $\Delta$  are %; Clause in words, Level in BAREC levels, Hard w. in SAMER words. Contra%: changed outputs with  $P(\text{contradict}) \geq 0.5$ . Arabic: Gemma’s “Arabic correct” (diagnostic). Unch.%: outputs identical to the input.

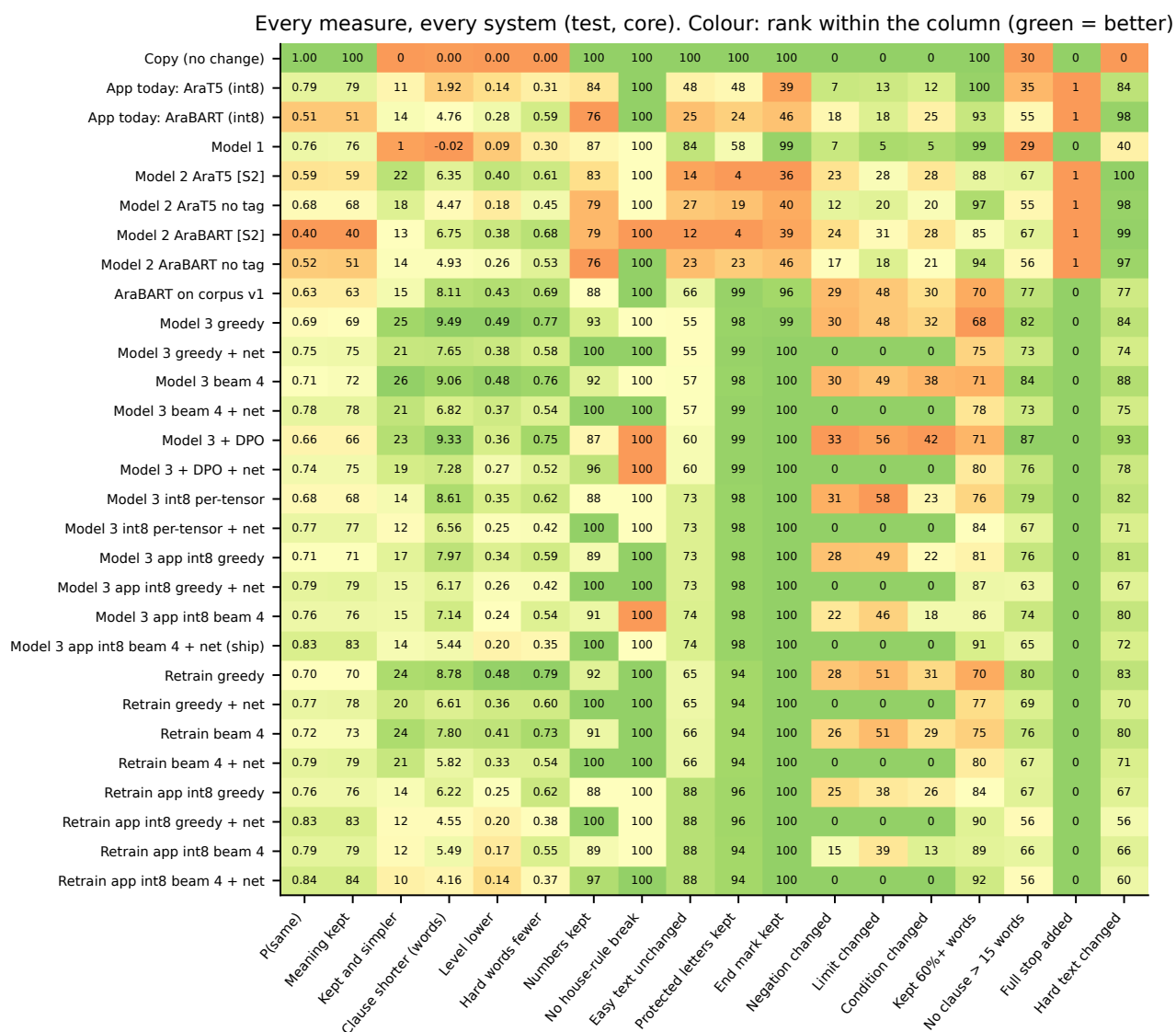
**Table 3:** Test split, core items.

| System                                      | P(same)     | Kept%     | Joint%    | Clause↓    | Level↓      | Hard w.↓    | Num.%      | Easy=%    | Prot.%    | Neg.Δ%   |
|---------------------------------------------|-------------|-----------|-----------|------------|-------------|-------------|------------|-----------|-----------|----------|
| Copy (no change)                            | 1.00        | 100       | 0         | 0.0        | 0.00        | 0.00        | 100        | 100       | 100       | 0        |
| App today: AraT5 (int8)                     | 0.79        | 79        | 11        | 1.9        | 0.14        | 0.31        | 84         | 48        | 48        | 7        |
| App today: AraBART (int8)                   | 0.51        | 51        | 14        | 4.8        | 0.28        | 0.59        | 76         | 25        | 24        | 18       |
| Model 1                                     | 0.76        | 76        | 1         | -0.0       | 0.09        | 0.30        | 87         | 84        | 58        | 7        |
| Model 2 AraT5 [S2]                          | 0.59        | 59        | 22        | 6.3        | 0.40        | 0.61        | 83         | 14        | 4         | 23       |
| Model 2 AraT5 no tag                        | 0.68        | 68        | 18        | 4.5        | 0.18        | 0.45        | 79         | 27        | 19        | 12       |
| Model 2 AraBART [S2]                        | 0.40        | 40        | 13        | 6.7        | 0.38        | 0.68        | 79         | 12        | 4         | 24       |
| Model 2 AraBART no tag                      | 0.52        | 51        | 14        | 4.9        | 0.26        | 0.53        | 76         | 23        | 23        | 17       |
| AraBART on corpus v1                        | 0.63        | 63        | 15        | 8.1        | 0.43        | 0.69        | 88         | 66        | <b>99</b> | 29       |
| Model 3 greedy                              | 0.69        | 69        | 25        | <b>9.5</b> | <b>0.49</b> | 0.77        | 93         | 55        | 98        | 30       |
| Model 3 greedy + net                        | 0.75        | 75        | 21        | 7.7        | 0.38        | 0.58        | <b>100</b> | 55        | <b>99</b> | <b>0</b> |
| Model 3 beam 4                              | 0.71        | 72        | <b>26</b> | 9.1        | 0.48        | 0.76        | 92         | 57        | 98        | 30       |
| Model 3 beam 4 + net                        | 0.78        | 78        | 21        | 6.8        | 0.37        | 0.54        | <b>100</b> | 57        | <b>99</b> | <b>0</b> |
| Model 3 + DPO                               | 0.66        | 66        | 23        | 9.3        | 0.36        | 0.75        | 87         | 60        | <b>99</b> | 33       |
| Model 3 + DPO + net                         | 0.74        | 75        | 19        | 7.3        | 0.27        | 0.52        | 96         | 60        | <b>99</b> | <b>0</b> |
| Model 3 int8 per-tensor                     | 0.68        | 68        | 14        | 8.6        | 0.35        | 0.62        | 88         | 73        | 98        | 31       |
| Model 3 int8 per-tensor + net               | 0.77        | 77        | 12        | 6.6        | 0.25        | 0.42        | <b>100</b> | 73        | 98        | <b>0</b> |
| Model 3 app int8 greedy                     | 0.71        | 71        | 17        | 8.0        | 0.34        | 0.59        | 89         | 73        | 98        | 28       |
| Model 3 app int8 greedy + net               | 0.79        | 79        | 15        | 6.2        | 0.26        | 0.42        | <b>100</b> | 73        | 98        | <b>0</b> |
| Model 3 app int8 beam 4                     | 0.76        | 76        | 15        | 7.1        | 0.24        | 0.54        | 91         | 74        | 98        | 22       |
| <b>Model 3 app int8 beam 4 + net (ship)</b> | <b>0.83</b> | <b>83</b> | 14        | 5.4        | 0.20        | 0.35        | <b>100</b> | 74        | 98        | <b>0</b> |
| Retrain greedy                              | 0.70        | 70        | 24        | 8.8        | 0.48        | <b>0.79</b> | 92         | 65        | 94        | 28       |
| Retrain greedy + net                        | 0.77        | 78        | 20        | 6.6        | 0.36        | 0.60        | <b>100</b> | 65        | 94        | <b>0</b> |
| Retrain beam 4                              | 0.72        | 73        | 24        | 7.8        | 0.41        | 0.73        | 91         | 66        | 94        | 26       |
| Retrain beam 4 + net                        | 0.79        | 79        | 21        | 5.8        | 0.33        | 0.54        | <b>100</b> | 66        | 94        | <b>0</b> |
| Retrain app int8 greedy                     | 0.76        | 76        | 14        | 6.2        | 0.25        | 0.62        | 88         | <b>88</b> | 96        | 25       |
| Retrain app int8 greedy + net               | 0.83        | 83        | 12        | 4.6        | 0.20        | 0.38        | <b>100</b> | <b>88</b> | 96        | <b>0</b> |
| Retrain app int8 beam 4                     | 0.79        | 79        | 12        | 5.5        | 0.17        | 0.55        | 89         | <b>88</b> | 94        | 15       |
| Retrain app int8 beam 4 + net               | <b>0.84</b> | <b>84</b> | 10        | 4.2        | 0.14        | 0.37        | 97         | <b>88</b> | 94        | <b>0</b> |
| Model 3 hybrid 0% beam                      | 0.75        | 75        | 21        | 7.7        | 0.38        | 0.58        | <b>100</b> | 55        | <b>99</b> | <b>0</b> |
| Model 3 hybrid 10% beam                     | 0.75        | 75        | 22        | 7.7        | 0.38        | 0.60        | <b>100</b> | 55        | <b>99</b> | <b>0</b> |
| Model 3 hybrid 20% beam                     | 0.76        | 76        | 22        | 7.6        | 0.38        | 0.57        | <b>100</b> | 55        | <b>99</b> | <b>0</b> |
| Model 3 hybrid 30% beam                     | 0.76        | 77        | 23        | 7.5        | 0.38        | 0.56        | <b>100</b> | 55        | <b>99</b> | <b>0</b> |
| Model 3 hybrid 50% beam                     | 0.77        | 78        | 23        | 7.1        | 0.37        | 0.55        | <b>100</b> | 56        | <b>99</b> | <b>0</b> |
| Model 3 hybrid 100% beam                    | 0.78        | 78        | 21        | 6.8        | 0.37        | 0.54        | <b>100</b> | 57        | <b>99</b> | <b>0</b> |



**Table 4:** Dev split, core items.

| System                                      | P(same)     | Kept%     | Joint%    | Clause↓    | Level↓      | Hard w.↓    | Num.%      | Easy=%    | Prot.%    | Neg.Δ%   |
|---------------------------------------------|-------------|-----------|-----------|------------|-------------|-------------|------------|-----------|-----------|----------|
| Copy (no change)                            | 1.00        | 100       | 0         | 0.0        | 0.00        | 0.00        | 100        | 100       | 100       | 0        |
| App today: AraT5 (int8)                     | 0.79        | 78        | 6         | 1.4        | 0.01        | 0.34        | 72         | 40        | 48        | 2        |
| App today: AraBART (int8)                   | 0.55        | 55        | 15        | 3.8        | 0.21        | 0.47        | 72         | 21        | 27        | 2        |
| Model 1                                     | 0.78        | 77        | 1         | 0.5        | -0.02       | 0.20        | 84         | 91        | 65        | 2        |
| Model 2 AraT5 [S2]                          | 0.63        | 62        | 25        | 6.4        | 0.20        | 0.58        | 76         | 19        | 12        | 20       |
| Model 2 AraT5 no tag                        | 0.78        | 78        | 17        | 3.7        | 0.00        | 0.41        | 72         | 26        | 35        | 6        |
| Model 2 AraBART [S2]                        | 0.42        | 42        | 12        | 6.0        | 0.25        | 0.46        | 72         | 13        | 7         | 18       |
| Model 2 AraBART no tag                      | 0.56        | 56        | 14        | 3.3        | 0.16        | 0.43        | 76         | 21        | 25        | 6        |
| AraBART on corpus v1                        | 0.62        | 62        | 15        | 7.9        | <b>0.48</b> | 0.58        | 84         | 64        | <b>95</b> | 20       |
| Model 3 greedy                              | 0.73        | 72        | <b>27</b> | 8.7        | 0.46        | 0.67        | 80         | 53        | 93        | 25       |
| Model 3 greedy + net                        | 0.77        | 78        | 23        | 6.7        | 0.33        | 0.57        | <b>100</b> | 53        | 93        | <b>0</b> |
| Model 3 beam 4                              | 0.76        | 76        | 26        | <b>8.8</b> | 0.45        | <b>0.78</b> | 84         | 64        | 93        | 28       |
| Model 3 beam 4 + net                        | 0.80        | 80        | 23        | 6.9        | 0.36        | 0.65        | <b>100</b> | 64        | <b>95</b> | <b>0</b> |
| Model 3 + DPO                               | 0.71        | 70        | 26        | 8.5        | 0.34        | 0.64        | 84         | 66        | 92        | 32       |
| Model 3 + DPO + net                         | 0.76        | 75        | 24        | 6.4        | 0.30        | 0.51        | <b>100</b> | 66        | 92        | <b>0</b> |
| Model 3 int8 per-tensor                     | 0.72        | 72        | 18        | 7.9        | 0.32        | 0.49        | 84         | 89        | 92        | 14       |
| Model 3 int8 per-tensor + net               | 0.76        | 76        | 15        | 6.8        | 0.26        | 0.39        | 96         | 89        | 92        | <b>0</b> |
| Model 3 app int8 greedy                     | 0.74        | 74        | 19        | 8.4        | 0.31        | 0.55        | 88         | 87        | 93        | 18       |
| Model 3 app int8 greedy + net               | 0.79        | 79        | 17        | 6.3        | 0.25        | 0.44        | <b>100</b> | 87        | 93        | <b>0</b> |
| Model 3 app int8 beam 4                     | 0.81        | 82        | 20        | 7.1        | 0.20        | 0.49        | 92         | 87        | 93        | 14       |
| <b>Model 3 app int8 beam 4 + net (ship)</b> | <b>0.85</b> | <b>85</b> | 18        | 5.4        | 0.16        | 0.37        | <b>100</b> | 87        | 93        | <b>0</b> |
| Retrain greedy                              | 0.73        | 74        | 21        | 7.3        | 0.39        | 0.60        | 88         | 74        | 92        | 24       |
| Retrain greedy + net                        | 0.79        | 79        | 18        | 5.2        | 0.30        | 0.51        | 96         | 74        | 92        | <b>0</b> |
| Retrain beam 4                              | 0.74        | 75        | 21        | 7.3        | 0.37        | 0.74        | 92         | 77        | 92        | 28       |
| Retrain beam 4 + net                        | 0.79        | 80        | 18        | 5.2        | 0.30        | 0.57        | <b>100</b> | 77        | 92        | <b>0</b> |
| Retrain app int8 greedy                     | 0.77        | 77        | 15        | 6.0        | 0.25        | 0.61        | 80         | 89        | 93        | 12       |
| Retrain app int8 greedy + net               | 0.80        | 80        | 14        | 4.8        | 0.22        | 0.52        | 92         | 89        | 93        | <b>0</b> |
| Retrain app int8 beam 4                     | 0.79        | 79        | 15        | 4.7        | 0.25        | 0.57        | 92         | <b>96</b> | 93        | 8        |
| Retrain app int8 beam 4 + net               | 0.83        | 82        | 15        | 4.0        | 0.22        | 0.49        | <b>100</b> | <b>96</b> | 93        | <b>0</b> |
| Model 3 hybrid 0% beam                      | 0.77        | 78        | 23        | 6.7        | 0.33        | 0.57        | <b>100</b> | 53        | 93        | <b>0</b> |
| Model 3 hybrid 10% beam                     | 0.76        | 77        | 24        | 7.3        | 0.35        | 0.64        | <b>100</b> | 53        | 92        | <b>0</b> |
| Model 3 hybrid 20% beam                     | 0.76        | 76        | 25        | 7.5        | 0.36        | 0.70        | <b>100</b> | 53        | 92        | <b>0</b> |
| Model 3 hybrid 30% beam                     | 0.76        | 77        | 25        | 7.5        | 0.39        | 0.68        | <b>100</b> | 55        | 92        | <b>0</b> |
| Model 3 hybrid 50% beam                     | 0.79        | 79        | 25        | 7.1        | 0.37        | 0.66        | <b>100</b> | 60        | 92        | <b>0</b> |
| Model 3 hybrid 100% beam                    | 0.80        | 80        | 23        | 6.9        | 0.36        | 0.65        | <b>100</b> | 64        | <b>95</b> | <b>0</b> |



**Figure 6:** Every measure for every system (test, core). Colour is the rank within the column, green better; numbers are the raw values.

## 6 Meaning

Meaning, test core items (dotted line: today's app)

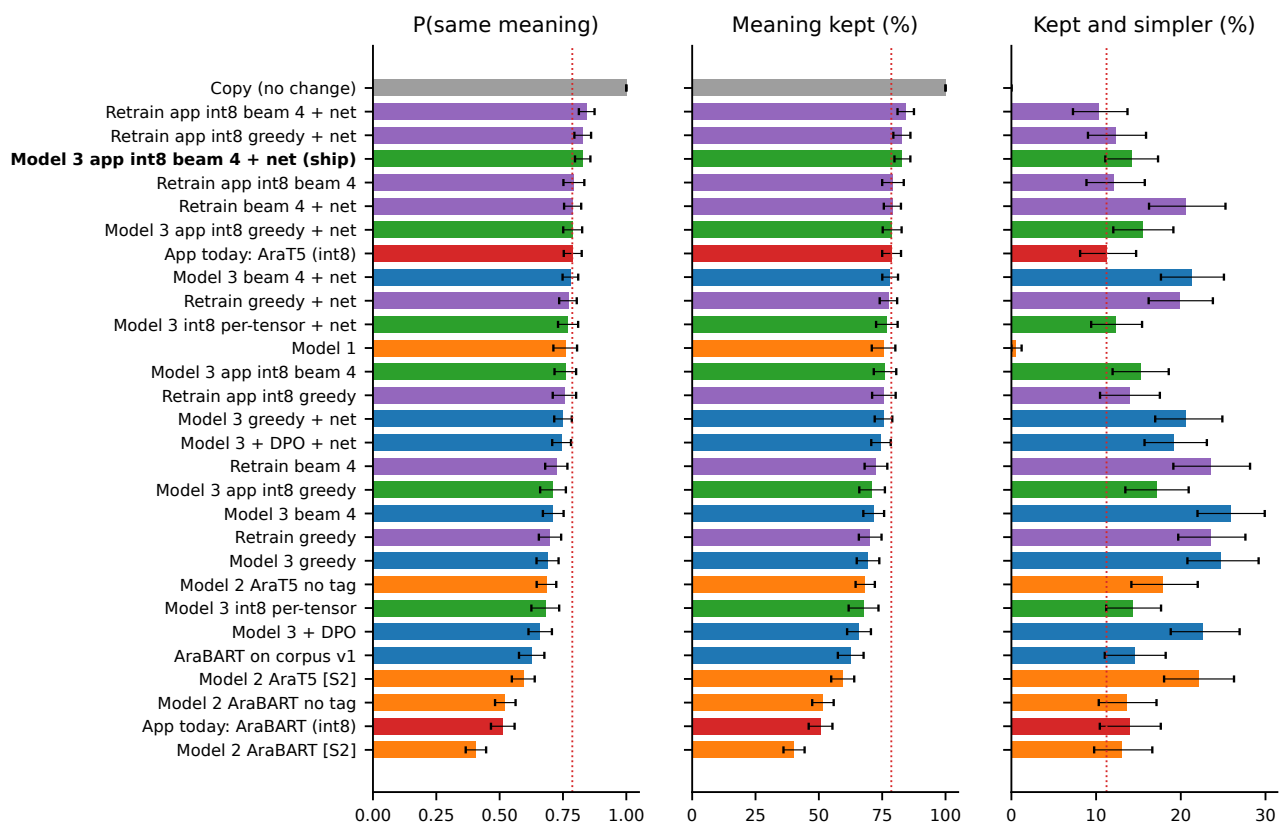


Figure 7: P(same), meaning kept and kept-and-simpler on test (core), all systems.

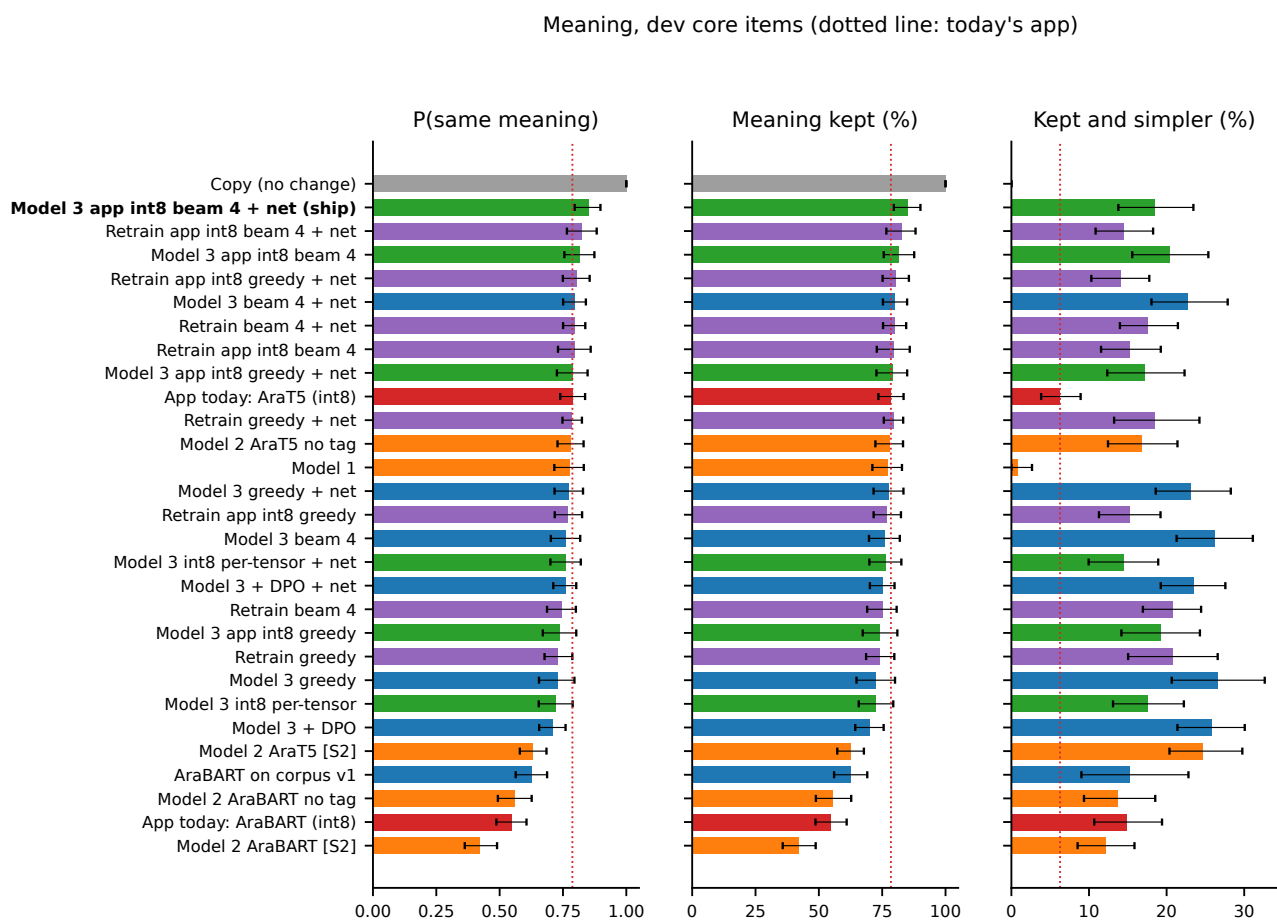


Figure 8: The same on dev.

Per-item P(same) on changed outputs only (test, core): copies excluded, so copying cannot inflate it

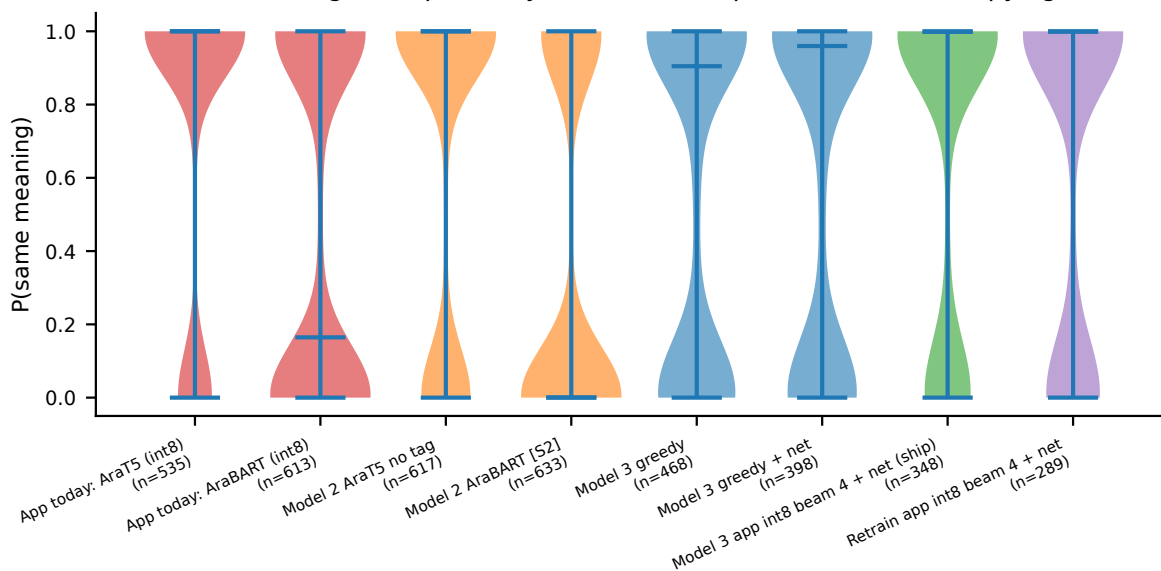
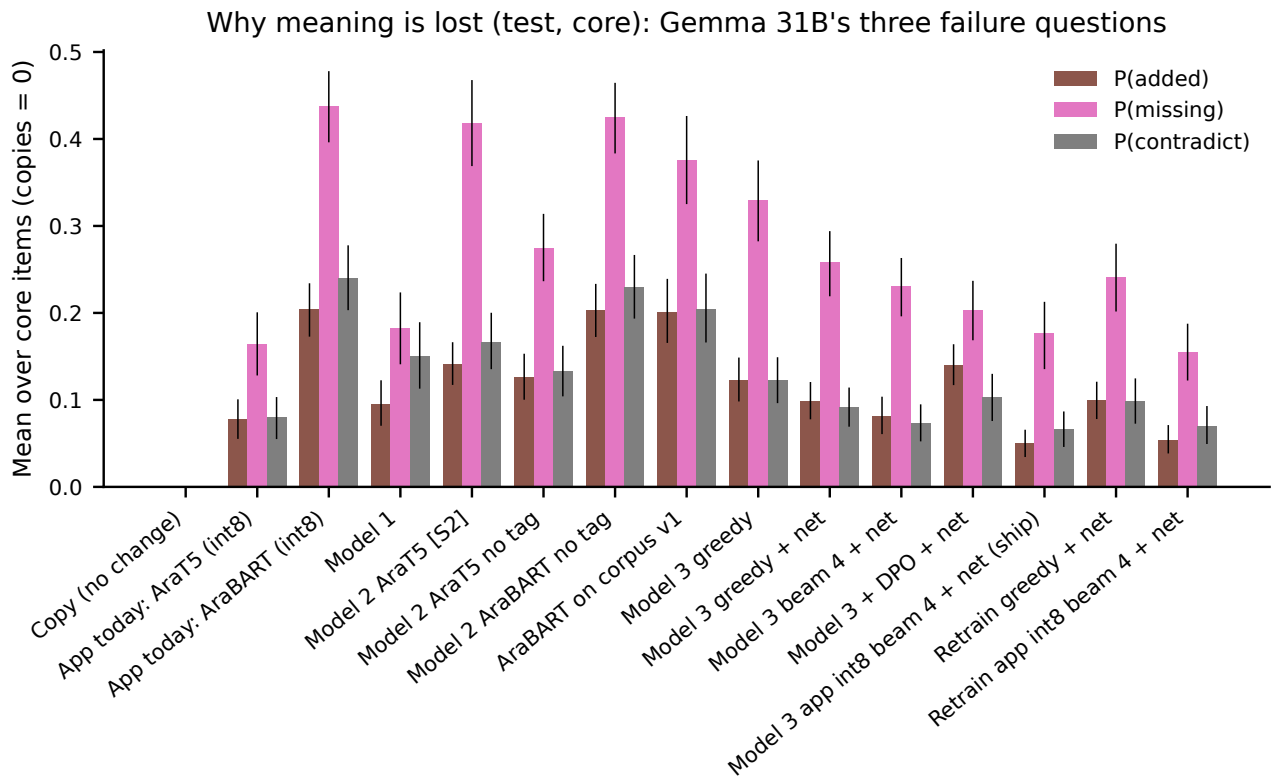
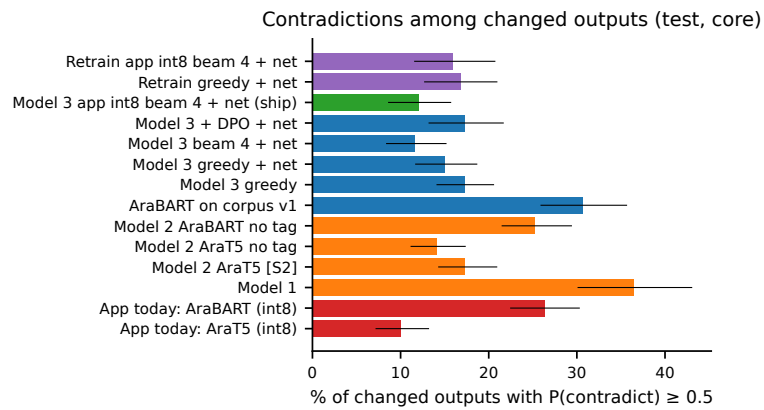


Figure 9: Distribution of P(same) over the outputs each system actually changed (copies excluded).

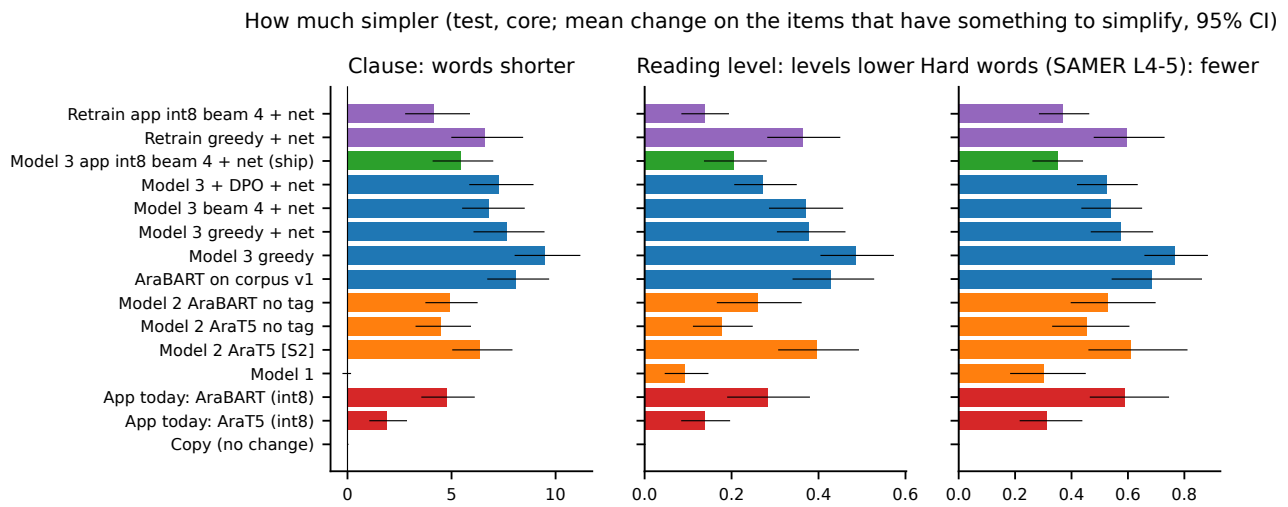


**Figure 10:** Mean P(added), P(missing) and P(contradict): when meaning is lost, it is mostly by leaving something out.



**Figure 11:** Share of changed outputs that Gemma reads as contradicting the source.

## 7 Simplicity



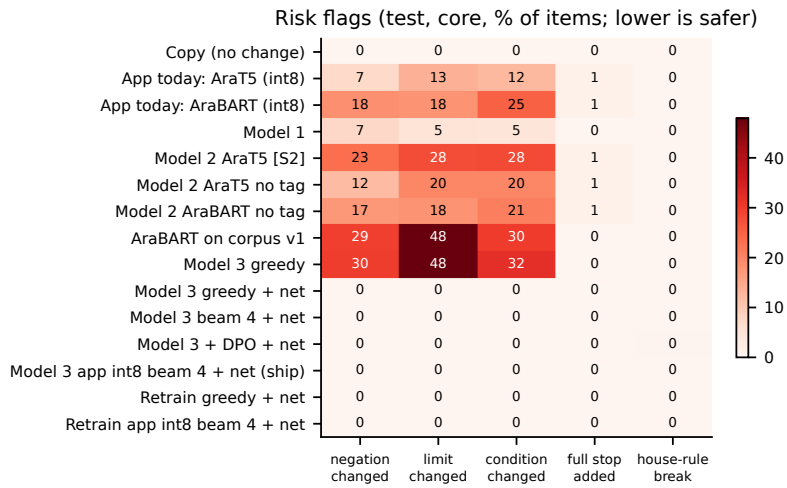
**Figure 12:** Clause shortening, reading-level drop and hard-word removal (test, core).

Float model 3 with the net shortens the longest clause by 7.7 words and lowers the reading level by 0.38; the shipping int8 bundle keeps most of the clause shortening (5.4) but only about half of the level drop (0.20). Today's app shortens clauses by 1.9 words and lowers the level by 0.14.

## 8 Safety and restraint

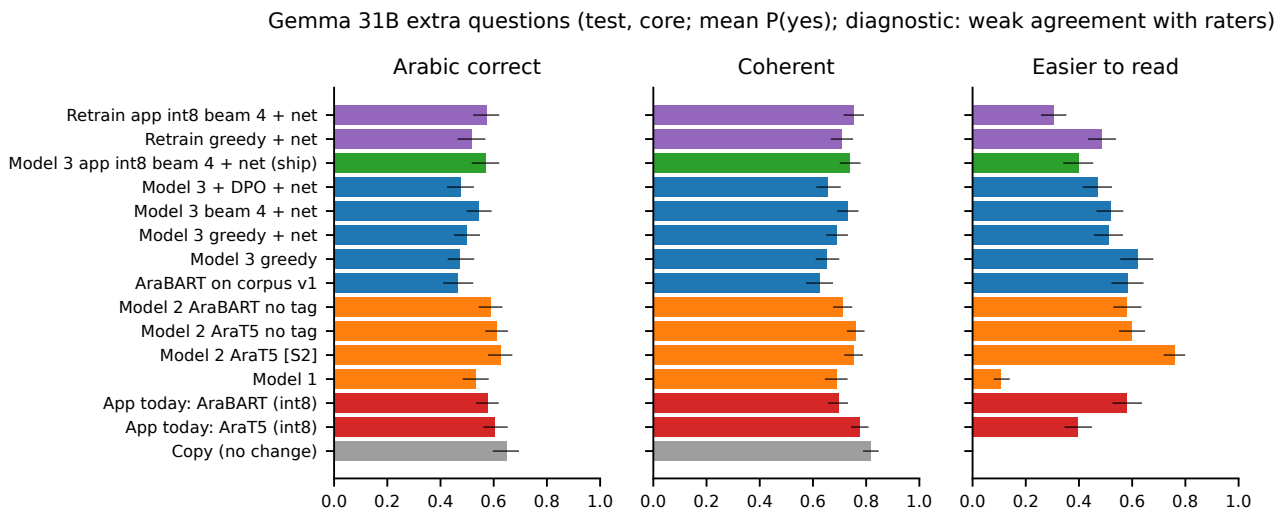


**Figure 13:** Numbers kept, easy text left unchanged (track C), protected text keeps its letters (track F) and cut-off endings keep their end mark (E1).



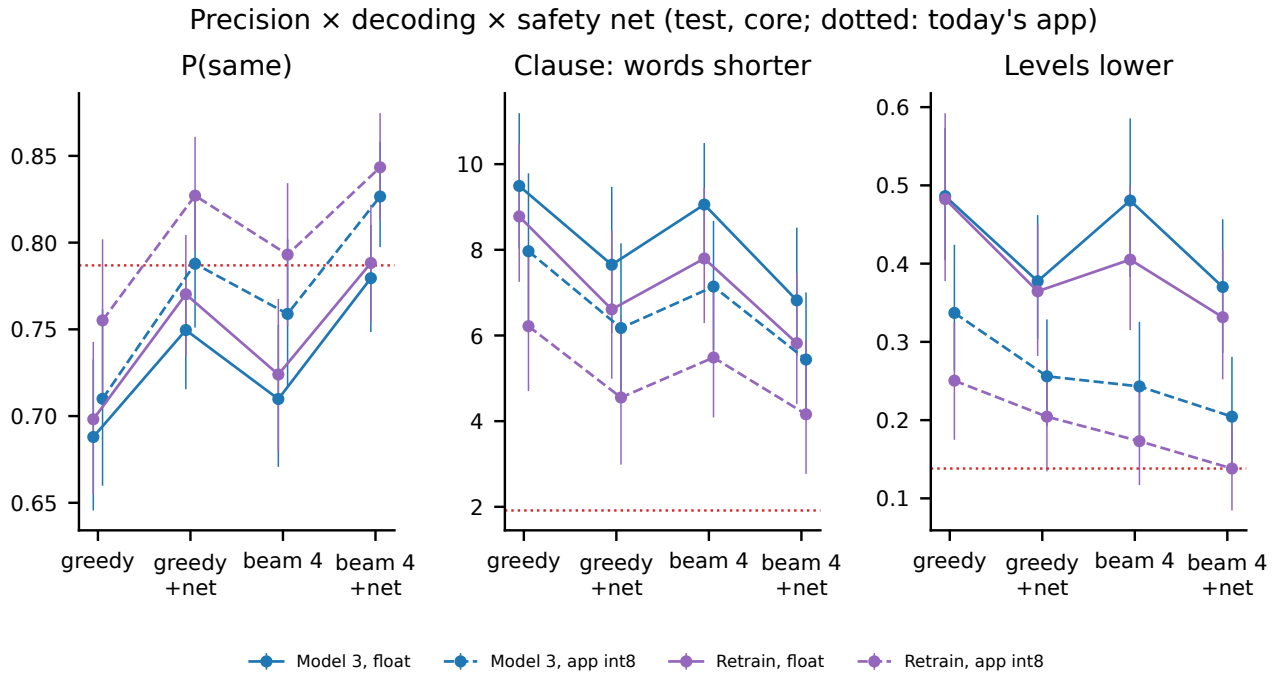
**Figure 14:** Risk flags: how often a system changes a negation, limit or condition word, adds a mid-sentence full stop, or breaks a house rule. The safety net brings the first three to zero by construction.

## 9 Arabic quality (diagnostic)

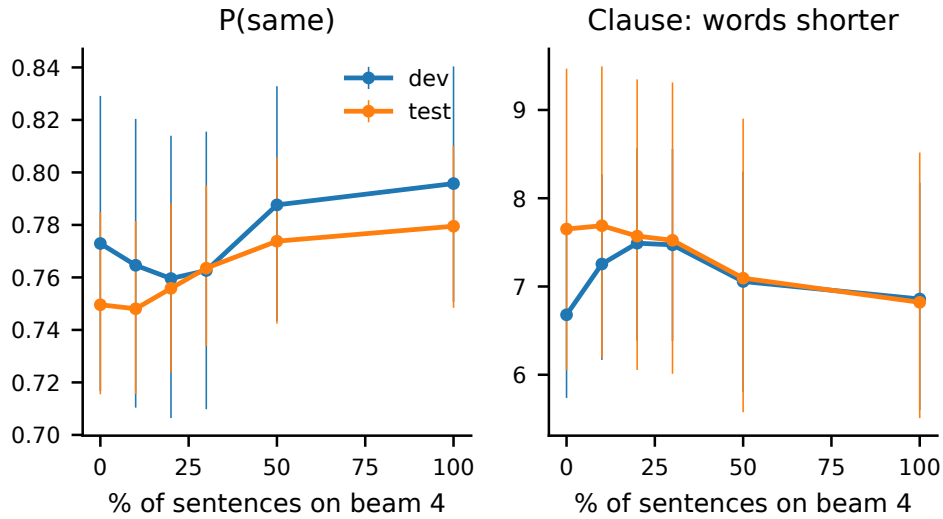


**Figure 15:** Gemma 31B's extra questions. Model 3's outputs are rated less correct and less coherent than the app's lighter edits; raters only weakly agree with these scores.

## 10 Precision, decoding and the safety net



**Figure 16:** Every combination of model (model 3, retrain), precision (float, app int8), decoding (greedy, beam 4) and safety net. Dotted: today's app.



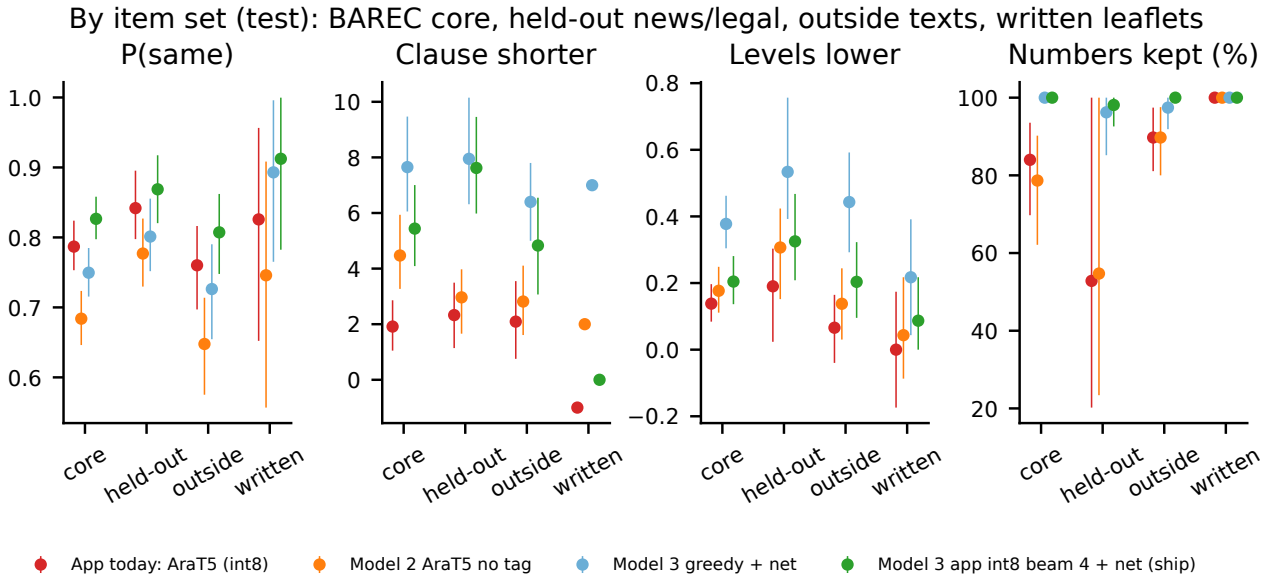
**Figure 17:** Hybrid decoding (float model 3): beam 4 on the least confident share of sentences. P(same) rises towards full beam 4 and clause shortening stays roughly flat; no partial share beats full beam 4.



**Table 5:** The app’s int8 recipe on two CPUs: exact matches with PyTorch greedy on reference sentences.

| CPU                             | int8 recipe                             | = PyTorch | Output                       |
|---------------------------------|-----------------------------------------|-----------|------------------------------|
| Xeon E5-2699 v4 (AVX2, no VNNI) | per-channel, MatMul+Gather (app recipe) | 0/12      | nonsense output (saturation) |
| Xeon E5-2699 v4 (AVX2, no VNNI) | per-channel, MatMul only                | 0/6       | nonsense output              |
| Xeon E5-2699 v4 (AVX2, no VNNI) | per-tensor                              | 1/6       | fluent, copies more          |
| Xeon E5-2699 v4 (AVX2, no VNNI) | per-channel, reduce_range               | 1/6       | fluent                       |
| i7-12700 (AVX-VNNI)             | per-channel, MatMul+Gather (app recipe) | 3/12      | fluent; 0.17 s/sentence      |

## 11 Item sets

**Figure 18:** Ship candidate vs today’s app, model 2 and float model 3 + net on each item set (test).

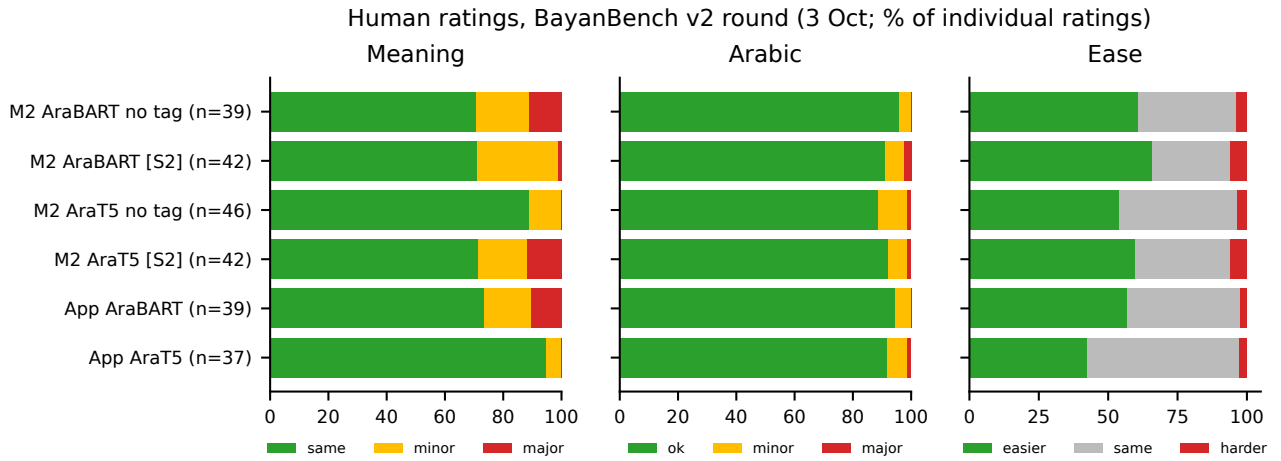
## 12 Reference-based scores

SARI against corpus v1 references and SAMER test (human L3 and L4 references), for the float systems scored in the all-systems comparison. Copy%: changed-reference rows the model leaves as a copy. Non-word%: output words not seen in BAREC train; Repeat%: outputs with a repeated word or 3-gram.

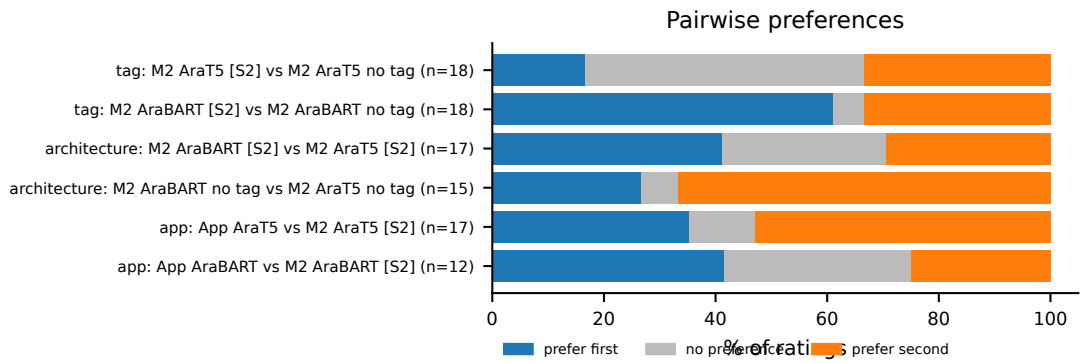
**Table 6:** Reference-based scores.

| System                 | Corpus v1 dev |       | Corpus v1 test |       | SAMER test (L3+L4 refs) |       | Quality, v1 test |         |
|------------------------|---------------|-------|----------------|-------|-------------------------|-------|------------------|---------|
|                        | SARI          | Copy% | SARI           | Copy% | SARI                    | Copy% | Non-word%        | Repeat% |
| Copy                   | 42.9          | 100   | 43.4           | 100   | 60.4                    | 100   | –                | –       |
| Model 1                | 41.6          | 53    | 41.4           | 53    | 54.9                    | 44    | 34.7             | 4.7     |
| Model 2 AraT5 [S2]     | 41.7          | 2     | 41.3           | 2     | 32.6                    | 1     | 10.7             | 2.2     |
| Model 2 AraT5 no tag   | 41.2          | 6     | 40.4           | 6     | 42.7                    | 10    | 10.4             | 2.7     |
| Model 2 AraBART [S2]   | 41.4          | 1     | 40.9           | 1     | 31.8                    | 2     | 11.5             | 1.9     |
| Model 2 AraBART no tag | 41.6          | 6     | 40.5           | 5     | 40.2                    | 8     | 13.3             | 2.8     |
| AraBART on corpus v1   | 51.8          | 15    | 52.8           | 15    | 42.9                    | 52    | 8.7              | 3.7     |
| Model 3 greedy         | 54.5          | 8     | 55.0           | 9     | 38.5                    | 40    | 9.4              | 3.5     |
| Model 3 + DPO          | 53.7          | 11    | 54.4           | 11    | 43.2                    | 52    | 9.4              | 6.8     |

## 13 Human ratings

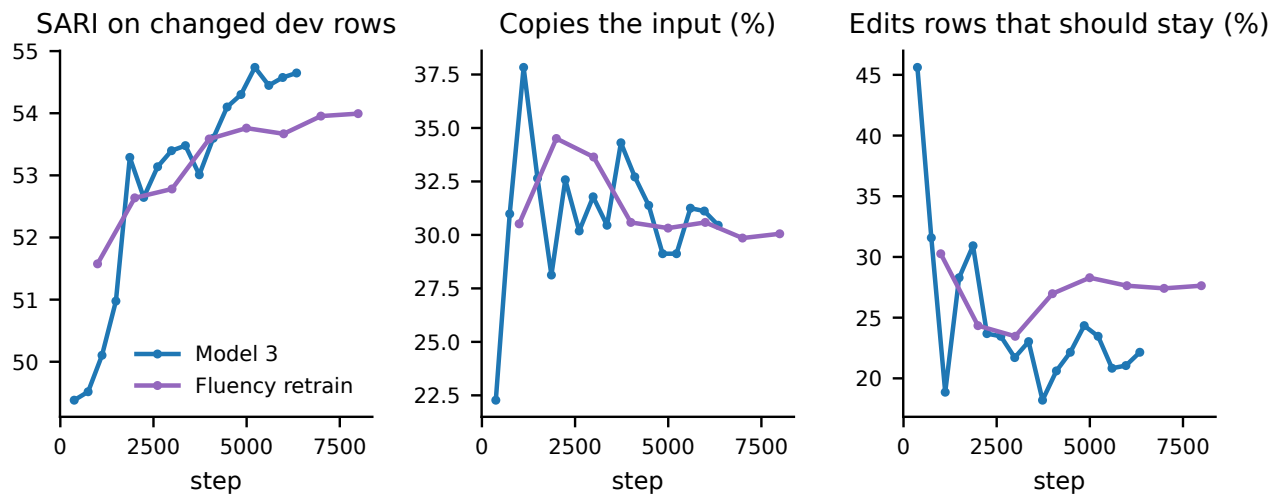


**Figure 19:** Human ratings of the systems in the rating round (model 3 was not yet built; the round is the basis of the scorer validation).

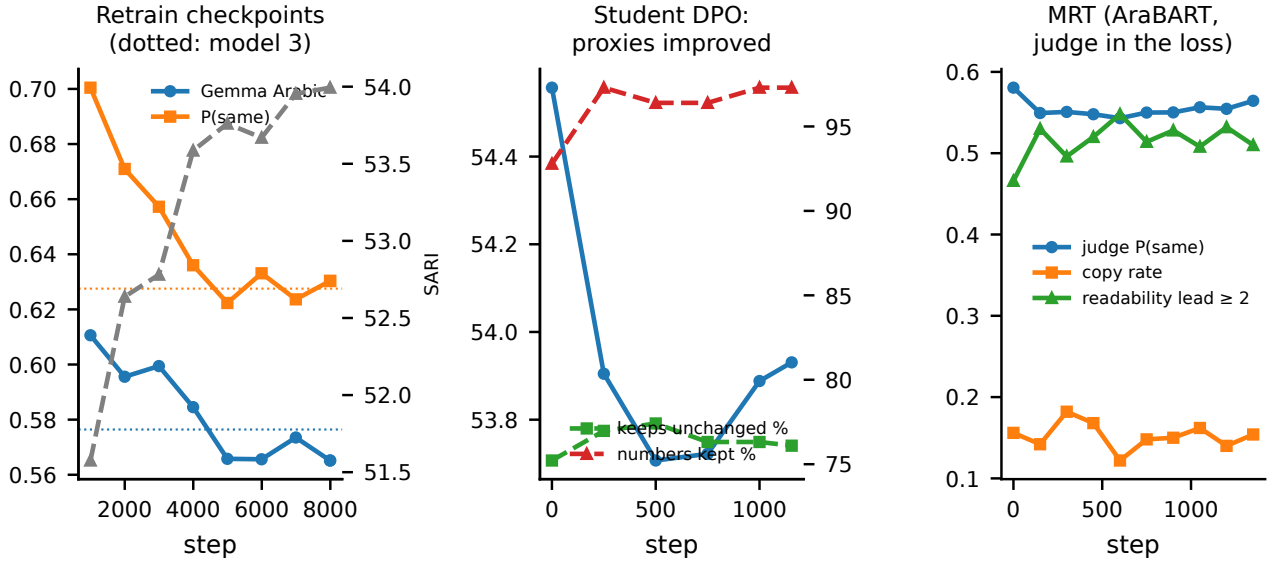


**Figure 20:** Side-by-side preferences: tag vs no tag, AraT5 vs AraBART, and the app vs model 2.

## 14 Training



**Figure 21:** Model 3 and the fluency retrain during training (dev evaluation): SARI on changed rows, copy rate and edits to rows that should stay unchanged.



**Figure 22:** Left: retrain checkpoints under the selection rule (pick the highest Gemma Arabic with SARI within 1 of the best and  $P(\text{same}) \geq \text{model 3's}$ ). Middle: student DPO improved every proxy but lost on Gemma 31B. Right: MRT with the judge in the loss.

**Table 7:** Retrain checkpoint selection (400 corpus v1 dev rows).

| Checkpoint | Gemma Arabic | P(same) | SARI |
|------------|--------------|---------|------|
| reference  | 0.576        | 0.628   | –    |
| step 999   | 0.611        | 0.700   | 51.6 |
| step 1998  | 0.596        | 0.671   | 52.6 |
| step 2997  | 0.599        | 0.657   | 52.8 |
| step 3996  | 0.585        | 0.636   | 53.6 |
| step 4995  | 0.566        | 0.622   | 53.8 |
| step 5994  | 0.566        | 0.633   | 53.7 |
| step 6993  | 0.574        | 0.624   | 54.0 |
| step 7992  | 0.565        | 0.630   | 54.0 |
| step 7996  | 0.565        | 0.630   | –    |

## 15 On-device cost

**Table 8:** Size and CPU latency. Hardware differs between rows (last column).

| Model                          | MB   | s/sentence | Hardware, decoding                                    |
|--------------------------------|------|------------|-------------------------------------------------------|
| Model 1 (PyTorch fp32)         | 1470 | 0.91       | vast.ai CPU, 4 threads, greedy                        |
| Model 2 AraT5 (PyTorch fp32)   | 1470 | 0.87       | vast.ai CPU, 4 threads, greedy                        |
| Model 2 AraBART (PyTorch fp32) | 557  | 0.42       | vast.ai CPU, 4 threads, greedy                        |
| Model 3 (PyTorch fp32)         | 1470 | 0.83       | vast.ai CPU, 4 threads, greedy                        |
| Model 3 ONNX fp32              | 1470 | 0.66       | vast.ai CPU, greedy                                   |
| Model 3 ONNX int8 per-tensor   | 740  | 0.45       | vast.ai CPU, greedy                                   |
| Model 3 app int8 bundle        | 471  | 0.17       | i7-12700 (AVX-VNNI), 4 threads, the app's greedy loop |

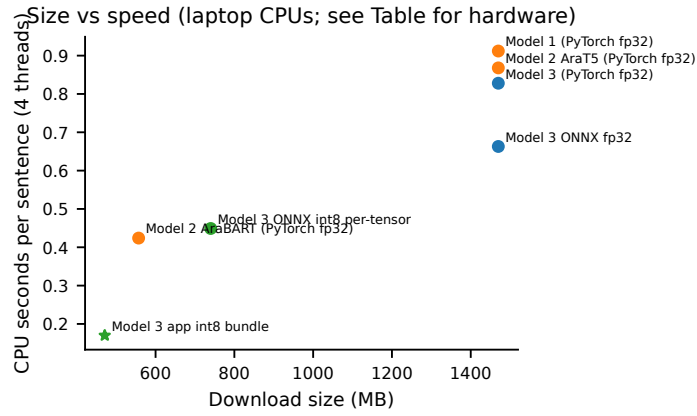


Figure 23: Size against CPU time per sentence.

Table 9: Model 3 artifacts.

| Hugging Face repo (private)           | Contents                                                                   | Size    |
|---------------------------------------|----------------------------------------------------------------------------|---------|
| Congi-libya/bayan-arat5-v1            | model 3, float weights                                                     | 1.47 GB |
| Congi-libya/bayan-arat5-v1-bundle     | model 3, app int8 bundle (encoder 170 MB, decoder 285 MB, tokenizer 15 MB) | 471 MB  |
| Congi-libya/bayan-arat5-v1-mix        | fluency retrain, float weights                                             | 1.47 GB |
| Congi-libya/bayan-arat5-v1-mix-bundle | fluency retrain, app int8 bundle                                           | 471 MB  |

## 16 Ship gate: the candidate against today's app

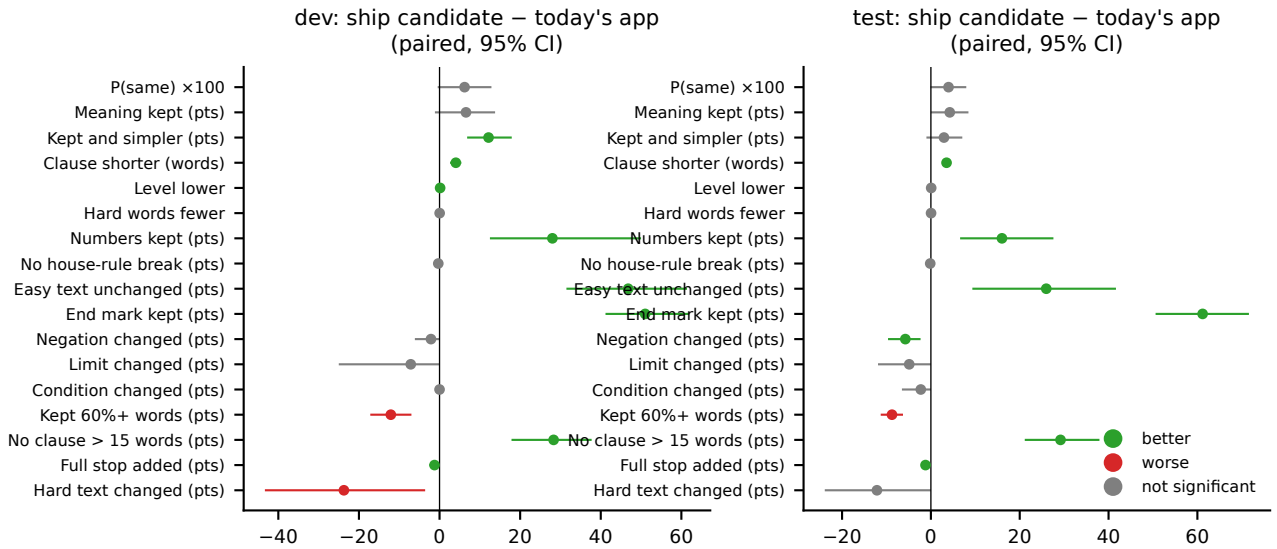


Figure 24: Paired differences, ship candidate minus today's app ( $P(\text{same})$  shown  $\times 100$ ). Green: significantly better; red: significantly worse; grey: interval includes zero.

| Measure (core items)   | App today | Ship candidate | Paired difference [95% CI]  |
|------------------------|-----------|----------------|-----------------------------|
| <i>dev split</i>       |           |                |                             |
| $P(\text{same})$       | 0.79      | 0.85           | +0.06 [-0.00, +0.13]        |
| Meaning kept           | 78        | 85             | +7 [-1, +14]                |
| Kept and simpler       | 6         | 18             | +12 [+7, +18] better        |
| Clause shorter (words) | 1.40      | 5.44           | +4.05 [+2.62, +5.39] better |
| Level lower            | 0.01      | 0.16           | +0.15 [+0.04, +0.26] better |

| Measure (core items)   | App today | Ship candidate | Paired difference [95% CI]         |
|------------------------|-----------|----------------|------------------------------------|
| Hard words fewer       | 0.34      | 0.37           | +0.03 [-0.12, +0.17]               |
| Numbers kept           | 72        | 100            | +28 [+12, +50] <b>better</b>       |
| No house-rule break    | 100       | 100            | -0 [-1, +0]                        |
| Easy text unchanged    | 40        | 87             | +47 [+31, +61] <b>better</b>       |
| Protected letters kept | 48        | 93             | +45 [+4, +73] <b>better</b>        |
| End mark kept          | 49        | 100            | +51 [+41, +62] <b>better</b>       |
| Negation changed       | 2         | 0              | -2 [-6, +0]                        |
| Limit changed          | 13        | 0              | -7 [-25, +0]                       |
| Condition changed      | 6         | 0              | +0 [+0, +0]                        |
| Kept 60%+ words        | 100       | 88             | -12 [-17, -7] <b>worse</b>         |
| No clause > 15 words   | 37        | 65             | +28 [+18, +38] <b>better</b>       |
| Full stop added        | 1         | 0              | -1 [-2, -0] <b>better</b>          |
| Hard text changed      | 85        | 61             | -24 [-43, -4] <b>worse</b>         |
| <i>test split</i>      |           |                |                                    |
| P(same)                | 0.79      | 0.83           | +0.04 [-0.00, +0.08]               |
| Meaning kept           | 79        | 83             | +4 [+0, +8]                        |
| Kept and simpler       | 11        | 14             | +3 [-1, +7]                        |
| Clause shorter (words) | 1.92      | 5.44           | +3.52 [+2.50, +4.54] <b>better</b> |
| Level lower            | 0.14      | 0.20           | +0.07 [-0.01, +0.15]               |
| Hard words fewer       | 0.31      | 0.35           | +0.04 [-0.10, +0.16]               |
| Numbers kept           | 84        | 100            | +16 [+7, +28] <b>better</b>        |
| No house-rule break    | 100       | 100            | -0 [-0, +0]                        |
| Easy text unchanged    | 48        | 74             | +26 [+9, +42] <b>better</b>        |
| Protected letters kept | 48        | 98             | +50 [+30, +67] <b>better</b>       |
| End mark kept          | 39        | 100            | +61 [+51, +72] <b>better</b>       |
| Negation changed       | 7         | 0              | -6 [-10, -2] <b>better</b>         |
| Limit changed          | 13        | 0              | -5 [-12, +0]                       |
| Condition changed      | 12        | 0              | -2 [-7, +0]                        |
| Kept 60%+ words        | 100       | 91             | -9 [-11, -6] <b>worse</b>          |
| No clause > 15 words   | 35        | 65             | +29 [+21, +38] <b>better</b>       |
| Full stop added        | 1         | 0              | -1 [-2, -0] <b>better</b>          |
| Hard text changed      | 84        | 72             | -12 [-24, +0]                      |

**Table 11:** Every BayanBench measure for the ship candidate and today’s app (core items), with the paired difference.

## 17 Examples

Test items, chosen at random within each category by fixed rules (`analyse.py`); scores are Gemma 31B’s.

### Good simplifications by the ship candidate

| B-7243b9ba (literature, track B): P(same) 1.00, Arabic 1.00, clause – words 6   |                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source                                                                          | والآن بعد أن كان ما كان، نعلم أنه رأى ظلًا ما على بنكرياسها، وهو عضو مدسوس بالقرب من الأمعاء ويفرز عصارة مليئةً بإنزيمات هاضمة والعديد من الهرمونات (منها الإنسولين).                                                           |
| Ship candidate                                                                  | والآن بعد أن كان ما كان، نعلم أنه رأى ظلًا ما على بنكرياسها، وهو عضو مدسوس قرب الأمعاء، ويفرز عصارة مليئةً بالإنزيمات الهاضمة والعديد من الهرمونات (منها الإنسولين).                                                            |
| App today                                                                       | وفي الوقت الحاضر بعد أن كان ما كان، نعلم أنه رأى ظلًا ما على بنكرياسها، وهو عضو مدسوس بالقرب من الأمعاء، ويفرز عصارة مليئةً بإنزيمات هاضمة وهرمونات (منها الإنسولين).                                                           |
| B-ae93affd (encyclopedia, track B): P(same) 1.00, Arabic 1.00, clause – words 6 |                                                                                                                                                                                                                                 |
| Source                                                                          | ومن أبرز الأحداث التي شهدتها لبنان في عهده: الحملة الفرنسية على مصر وفلسطين خلال عهد الوالي أحمد باشا الجزار، والحملة المصرية على بلاد الشام والمعارك العديدة التي خاضها الجيش المصري مع الجيش التركي للسيطرة على سوريا الكبرى. |

|                |                                                                                                                                                                                                                                 |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ship candidate | ومن أهم أحداث لبنان في عهده: الحملة الفرنسية على مصر وفلسطين في عهد الوالي أحمد باشا الجزار، والحملة المصرية على بلاد الشام، ومعارك كثيرة خاضها الجيش المصري مع الجيش التركي للسيطرة على سوريا الكبرى.                          |
| App today      | ومن أبرز الأحداث التي شهدتها لبنان في عهده: الحملة الفرنسية على مصر وفلسطين خلال عهد الوالي أحمد باشا الجزار، والحملة المصرية على بلاد الشام والمعارك العديدة التي خاضها الجيش المصري مع الجيش التركي للسيطرة على سوريا الكبرى. |

### Safety-net catches (the plain output was replaced by the source)

| B-d8eadf1a (children, track B): P(same) 1.00, Arabic 0.02, clause —words 0 |                                                                                                                                                                                                  |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source                                                                     | إذا كنت ترغب في تدوين ملاحظاتك وإجراء بعض العمليات الحسابية بواسطة القلم، والرسم وعمل مخططات بسيطة وسريعة، فإن اللوح الكمبيوتر الذي أطلق عليه اسم "نوت سلات" Note سوف Slate يكون مفيداً جداً لك. |
| Model output                                                               | يمكنك كتابة ملاحظاتك بالقلم والرسم وعمل مخططات بسيطة وسريعة، هذا اللوح اسمه نوت سلات Note سيكون مفيداً جداً لك.                                                                                  |
| Shown (net)                                                                | إذا كنت ترغب في تدوين ملاحظاتك وإجراء بعض العمليات الحسابية بواسطة القلم، والرسم وعمل مخططات بسيطة وسريعة، فإن اللوح الكمبيوتر الذي أطلق عليه اسم "نوت سلات" Note سوف Slate يكون مفيداً جداً لك. |
| App today                                                                  | إذا أردت كتابة ملاحظاتك وإجراء بعض العمليات الحسابية باستخدام القلم والرسم وعمل المخططات البسيطة، فإن اللوح الكمبيوتر الذي أطلق عليه اسم "نوت سلات" سيكون مفيداً جداً لك.                        |

| A1-87d8bc4d (children, track A1): P(same) 1.00, Arabic 0.05, clause —words 0 |                                                                                                                                                  |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Source                                                                       | وفضلاً عما يسببه ذلك من تشوه لمظهر الأظافر ويحول على المدى الطويل دون نموه بالشكل الطبيعي فإنه يمنحك مظهراً عاماً سيئاً وخاصة في الأماكن العامة، |
| Model output                                                                 | هذا يسبب تشوه في شكل الأظافر، وهو يمنع نموه بشكل طبيعي، وهذا يسبب لك مظهراً سيئاً في الأماكن العامة،                                             |
| Shown (net)                                                                  | وفضلاً عما يسببه ذلك من تشوه لمظهر الأظافر ويحول على المدى الطويل دون نموه بالشكل الطبيعي فإنه يمنحك مظهراً عاماً سيئاً وخاصة في الأماكن العامة، |
| App today                                                                    | وفي المقابل، يسبب ذلك تشوه لمظهر الأظافر ويحول على المدى الطويل دون نموه الطبيعي، فهو يمنحك مظهراً عاماً سيئاً خاصة في الأماكن العامة.           |

### Restraint: easy text the app changed and the ship candidate left alone

| C-f2fce558 (outside-digital, track C): P(same) 1.00, Arabic 0.00 |                                                     |
|------------------------------------------------------------------|-----------------------------------------------------|
| Source                                                           | انقر على التطبيقات التي تحققت من حالة حماية الجهاز. |
| Ship candidate                                                   | انقر على التطبيقات التي تحققت من حالة حماية الجهاز. |
| App today                                                        | انقر على التطبيقات التي أظهرت أنك تحمي جهازك.       |

| C-bc84be82 (children, track C): P(same) 1.00, Arabic 0.01 |                                                  |
|-----------------------------------------------------------|--------------------------------------------------|
| Source                                                    | من هناك طلب شراء كيس كبير من حبوب القمر اللذيذة، |
| Ship candidate                                            | من هناك طلب شراء كيس كبير من حبوب القمر اللذيذة، |
| App today                                                 | من يطلب شراء كيس كبير من حبوب القمر اللذيذة،     |

## Meaning slips (Gemma P(same) < 0.2)

| E2-50b4428d (encyclopedia, track E2): P(same) 0.00, Arabic 0.00 |                                                                                                                                           |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Source                                                          | بحسب تقرير مجلة فوربس يُعتبر إنتر ميلان من أغنى أندية العالم، الثالث في إيطاليا، حيث بلغت قيمته السوقية لسنة 2015 مبلغ 439 مليون دولار،   |
| Ship candidate                                                  | بحسب تقرير مجلة فوربس يعتبر إنتر ميلان من أغنى الأندية العالم الثالث في إيطاليا، حيث بلغت قيمته السوقية لسنة 2015 مبلغ 439 مليون دولار،   |
| App today                                                       | حسب تقرير مجلة فوربس، يعتبر إنتر ميلان من أغنى الأندية. العالم، الثالث في إيطاليا، حيث بلغت قيمته السوقية لسنة 2015 مبلغ 439 مليون دولار. |

| B-781afbfa (outside-health, track B): P(same) 0.01, Arabic 1.00, clause – words 10 |                                                                                                                                                                   |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source                                                                             | ويقدم تحالف غافي الدعم لحملات التطعيم التداركي ضد التهاب الدماغ الياباني ويشارك في تمويل اللقاحات اللازمة لجهود التمنيع الروتيني في البلدان المؤهلة للحصول عليها. |
| Ship candidate                                                                     | دعم تحالف غافي حملات التطعيم ضد التهاب الدماغ الياباني، كما ساعد في تمويل اللقاحات اللازمة لجهود التمنيع الروتيني في البلدان التي تحتاج إليها.                    |
| App today                                                                          | يقدم تحالف غافي الدعم لحملات التطعيم التداركي ضد التهاب الدماغ الياباني، ويشارك في تمويل اللقاحات اللازمة لجهود التمنيع الروتيني في البلدان المؤهلة للحصول عليها. |

## Grammar problems (Gemma “Arabic correct” < 0.2, meaning kept)

| A3-ff4e57a7 (outside-health, track A3): P(same) 0.75, Arabic 0.02 |                                                                                                                   |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Source                                                            | لا يوجد علاج لهذا المرض، ويركز العلاج على تخفيف الأعراض السريرية الوخيمة وتزويد المريض بالدعم اللازم لدرء العدوى. |
| Ship candidate                                                    | لا يوجد علاج لهذا المرض، العلاج يركز على تخفيف أعراض المرض، ويعطي المريض الدعم اللازم لكي يتخلص من العدوى.        |
| App today                                                         | لا يوجد علاج لهذا المرض، يركز العلاج على تخفيف الأعراض السريرية ومساعدة المريض على التخلص من العدوى.              |

| A1-e483d119 (learner, track A1): P(same) 1.00, Arabic 0.01 |                                                    |
|------------------------------------------------------------|----------------------------------------------------|
| Source                                                     | هيا انهضن كي لا تتأخر فنحن ذاهبون إلى المطار!      |
| Ship candidate                                             | هيا انهضن لكي لا تتأخر فنحن ذاهبون إلى المطار!     |
| App today                                                  | هيا هيا اركضوا كي لا تتأخر فنحن ذاهبون إلى المطار! |

## 18 Limitations

- **One scorer.** Meaning rests on Gemma 4 31B (AUC 0.85 against humans); the second-judge pilot failed, so there is no independent check yet. The extra questions are diagnostic only.
- **Copies count as perfect meaning.** Systems that copy more look more faithful; we report unchanged rates and P(same) on changed outputs alongside.
- **The joint rate uses our own minimum change** (2 clause words or half a level); v2.0 leaves it open.
- **Human ratings predate model 3.** The rating round covered the app and model 2 only.
- **int8 was checked on two x86 CPUs, not on a phone.** ARM uses the same integer dot-product path as VNNI, but a phone latency and output check is still needed.
- **Latency numbers come from different machines** (Table 8).

## A Per-set leaderboards

**Table 12:** Test, held-out items (news, legal).

| System                                      | P(same)     | Kept%     | Joint%    | Clause↓     | Level↓      | Hard w.↓    | Num.%      | Easy=%    | Prot.% | Neg.Δ%   |
|---------------------------------------------|-------------|-----------|-----------|-------------|-------------|-------------|------------|-----------|--------|----------|
| Copy (no change)                            | 1.00        | 100       | 0         | 0.0         | 0.00        | 0.00        | 100        | 100       | –      | 0        |
| App today: AraT5 (int8)                     | 0.84        | 77        | 8         | 2.3         | 0.19        | 0.33        | 53         | 23        | –      | 14       |
| App today: AraBART (int8)                   | 0.61        | 57        | 13        | 3.9         | 0.38        | 0.40        | 51         | 18        | –      | 21       |
| Model 1                                     | 0.83        | 83        | 1         | 0.4         | 0.03        | 0.28        | 96         | <b>87</b> | –      | 4        |
| Model 2 AraT5 [S2]                          | 0.71        | 65        | 19        | 5.8         | 0.42        | 0.47        | 53         | 7         | –      | 23       |
| Model 2 AraT5 no tag                        | 0.78        | 71        | 15        | 3.0         | 0.31        | 0.37        | 55         | 15        | –      | 19       |
| Model 2 AraBART [S2]                        | 0.52        | 49        | 14        | 5.5         | 0.50        | 0.45        | 55         | 8         | –      | 21       |
| Model 2 AraBART no tag                      | 0.63        | 58        | 13        | 3.9         | 0.40        | 0.44        | 53         | 20        | –      | 19       |
| AraBART on corpus v1                        | 0.67        | 66        | 13        | 10.3        | 0.67        | 0.56        | 92         | 49        | –      | 17       |
| Model 3 greedy                              | 0.74        | 74        | 23        | 10.8        | <b>0.71</b> | 0.57        | 89         | 36        | –      | 30       |
| Model 3 greedy + net                        | 0.80        | 80        | 18        | 7.9         | 0.53        | 0.52        | 96         | 41        | –      | <b>0</b> |
| Model 3 beam 4                              | 0.79        | 78        | <b>25</b> | 10.3        | 0.60        | 0.48        | 94         | 43        | –      | 25       |
| Model 3 beam 4 + net                        | 0.83        | 83        | 21        | 7.6         | 0.48        | 0.42        | 98         | 48        | –      | <b>0</b> |
| Model 3 + DPO                               | 0.75        | 74        | <b>25</b> | 10.1        | 0.49        | 0.42        | 94         | 48        | –      | 33       |
| Model 3 + DPO + net                         | 0.81        | 80        | 20        | 7.4         | 0.38        | 0.38        | 96         | 53        | –      | <b>0</b> |
| Model 3 int8 per-tensor                     | 0.77        | 77        | 18        | 10.3        | 0.49        | 0.43        | 92         | 59        | –      | 22       |
| Model 3 int8 per-tensor + net               | 0.84        | 84        | 17        | 7.6         | 0.39        | 0.36        | <b>100</b> | 63        | –      | <b>0</b> |
| Model 3 app int8 greedy                     | 0.76        | 77        | 17        | 10.6        | 0.45        | 0.44        | 94         | 57        | –      | 18       |
| Model 3 app int8 greedy + net               | 0.82        | 82        | 14        | 7.9         | 0.36        | 0.36        | <b>100</b> | 62        | –      | <b>0</b> |
| Model 3 app int8 beam 4                     | 0.84        | 85        | 18        | 9.1         | 0.35        | 0.31        | 94         | 65        | –      | 14       |
| <b>Model 3 app int8 beam 4 + net (ship)</b> | <b>0.87</b> | <b>88</b> | 16        | 7.6         | 0.33        | 0.26        | 98         | 68        | –      | <b>0</b> |
| Retrain greedy                              | 0.75        | 75        | 22        | <b>11.2</b> | 0.58        | 0.64        | 89         | 50        | –      | 21       |
| Retrain greedy + net                        | 0.81        | 81        | 20        | 7.9         | 0.48        | 0.53        | 98         | 54        | –      | <b>0</b> |
| Retrain beam 4                              | 0.80        | 80        | 23        | 10.8        | 0.58        | <b>0.66</b> | 94         | 54        | –      | 27       |
| Retrain beam 4 + net                        | 0.85        | 85        | 21        | 7.8         | 0.48        | 0.57        | <b>100</b> | 60        | –      | <b>0</b> |
| Retrain app int8 greedy                     | 0.80        | 81        | 17        | 8.8         | 0.42        | 0.33        | 91         | 70        | –      | 11       |
| Retrain app int8 greedy + net               | 0.84        | 85        | 15        | 6.8         | 0.37        | 0.30        | <b>100</b> | 72        | –      | <b>0</b> |
| Retrain app int8 beam 4                     | 0.84        | 84        | 13        | 7.6         | 0.36        | 0.40        | 92         | 73        | –      | 12       |
| Retrain app int8 beam 4 + net               | <b>0.87</b> | 87        | 13        | 5.8         | 0.30        | 0.35        | <b>100</b> | 76        | –      | <b>0</b> |
| Model 3 hybrid 0% beam                      | 0.80        | 80        | 18        | 7.9         | 0.53        | 0.52        | 96         | 41        | –      | <b>0</b> |
| Model 3 hybrid 10% beam                     | 0.80        | 80        | 18        | 7.9         | 0.53        | 0.49        | 96         | 42        | –      | <b>0</b> |
| Model 3 hybrid 20% beam                     | 0.82        | 82        | 19        | 7.8         | 0.47        | 0.46        | 96         | 43        | –      | <b>0</b> |
| Model 3 hybrid 30% beam                     | 0.81        | 81        | 18        | 7.7         | 0.47        | 0.45        | 98         | 45        | –      | <b>0</b> |
| Model 3 hybrid 50% beam                     | 0.83        | 82        | 21        | 7.6         | 0.49        | 0.43        | 98         | 48        | –      | <b>0</b> |
| Model 3 hybrid 100% beam                    | 0.83        | 83        | 21        | 7.6         | 0.48        | 0.42        | 98         | 48        | –      | <b>0</b> |



Table 13: Test, outside items.

| System                                      | P(same)     | Kept%     | Joint%    | Clause↓    | Level↓      | Hard w.↓    | Num.%      | Easy=%    | Prot.% | Neg.Δ%   |
|---------------------------------------------|-------------|-----------|-----------|------------|-------------|-------------|------------|-----------|--------|----------|
| Copy (no change)                            | 1.00        | 100       | 0         | 0.0        | 0.00        | 0.00        | 100        | 100       | –      | 0        |
| App today: AraT5 (int8)                     | 0.76        | 75        | 11        | 2.1        | 0.07        | 0.20        | 90         | 27        | –      | 21       |
| App today: AraBART (int8)                   | 0.46        | 47        | 8         | 4.2        | 0.15        | 0.48        | 87         | 12        | –      | 23       |
| Model 1                                     | <b>0.84</b> | <b>84</b> | 1         | 0.4        | 0.06        | 0.08        | 97         | <b>91</b> | –      | 10       |
| Model 2 AraT5 [S2]                          | 0.60        | 61        | 15        | 3.5        | 0.24        | 0.43        | 95         | 6         | –      | 38       |
| Model 2 AraT5 no tag                        | 0.65        | 64        | 15        | 2.8        | 0.14        | 0.41        | 90         | 12        | –      | 29       |
| Model 2 AraBART [S2]                        | 0.39        | 38        | 7         | 4.9        | 0.29        | 0.53        | 90         | 21        | –      | 25       |
| Model 2 AraBART no tag                      | 0.48        | 48        | 11        | 3.9        | 0.16        | 0.51        | 87         | 12        | –      | 26       |
| AraBART on corpus v1                        | 0.50        | 51        | 14        | 8.3        | <b>0.62</b> | 0.76        | 87         | 30        | –      | 30       |
| Model 3 greedy                              | 0.63        | 62        | 26        | 9.1        | 0.59        | 0.79        | 87         | 27        | –      | 37       |
| Model 3 greedy + net                        | 0.73        | 72        | 22        | 6.4        | 0.44        | 0.63        | 97         | 30        | –      | <b>0</b> |
| Model 3 beam 4                              | 0.65        | 64        | 28        | 8.8        | 0.48        | 0.70        | 92         | 45        | –      | 31       |
| Model 3 beam 4 + net                        | 0.74        | 73        | 24        | 7.0        | 0.41        | 0.56        | 97         | 45        | –      | <b>0</b> |
| Model 3 + DPO                               | 0.57        | 56        | 23        | 9.0        | 0.46        | 0.73        | 77         | 45        | –      | 41       |
| Model 3 + DPO + net                         | 0.66        | 66        | 19        | 7.5        | 0.38        | 0.59        | 90         | 48        | –      | <b>0</b> |
| Model 3 int8 per-tensor                     | 0.64        | 63        | 15        | 8.4        | 0.43        | 0.58        | 85         | 58        | –      | 35       |
| Model 3 int8 per-tensor + net               | 0.75        | 75        | 13        | 5.8        | 0.32        | 0.42        | 97         | 61        | –      | <b>0</b> |
| Model 3 app int8 greedy                     | 0.66        | 67        | 19        | 7.4        | 0.32        | 0.53        | 87         | 64        | –      | 39       |
| Model 3 app int8 greedy + net               | 0.74        | 75        | 16        | 5.5        | 0.29        | 0.41        | 97         | 67        | –      | <b>0</b> |
| Model 3 app int8 beam 4                     | 0.75        | 76        | 17        | 6.5        | 0.22        | 0.48        | 97         | 70        | –      | 27       |
| <b>Model 3 app int8 beam 4 + net (ship)</b> | 0.81        | 81        | 15        | 4.8        | 0.20        | 0.37        | <b>100</b> | 73        | –      | <b>0</b> |
| Retrain greedy                              | 0.58        | 59        | 26        | <b>9.4</b> | 0.58        | <b>0.82</b> | 82         | 48        | –      | 42       |
| Retrain greedy + net                        | 0.74        | 75        | 21        | 6.8        | 0.41        | 0.60        | 97         | 52        | –      | <b>0</b> |
| Retrain beam 4                              | 0.68        | 68        | <b>29</b> | 9.0        | 0.47        | 0.78        | 85         | 58        | –      | 32       |
| Retrain beam 4 + net                        | 0.79        | 79        | 25        | 6.8        | 0.37        | 0.66        | 97         | 61        | –      | <b>0</b> |
| Retrain app int8 greedy                     | 0.59        | 59        | 18        | 8.2        | 0.49        | 0.66        | 87         | 67        | –      | 37       |
| Retrain app int8 greedy + net               | 0.72        | 72        | 16        | 6.0        | 0.38        | 0.52        | <b>100</b> | 73        | –      | <b>0</b> |
| Retrain app int8 beam 4                     | 0.74        | 73        | 20        | 6.7        | 0.37        | 0.53        | 82         | 70        | –      | 33       |
| Retrain app int8 beam 4 + net               | 0.82        | 82        | 17        | 4.9        | 0.34        | 0.41        | 95         | 73        | –      | <b>0</b> |
| Model 3 hybrid 0% beam                      | 0.73        | 72        | 22        | 6.4        | 0.44        | 0.63        | 97         | 30        | –      | <b>0</b> |
| Model 3 hybrid 10% beam                     | 0.73        | 73        | 23        | 6.5        | 0.45        | 0.63        | 97         | 39        | –      | <b>0</b> |
| Model 3 hybrid 20% beam                     | 0.74        | 74        | 25        | 7.2        | 0.46        | 0.62        | 97         | 39        | –      | <b>0</b> |
| Model 3 hybrid 30% beam                     | 0.75        | 75        | 26        | 7.3        | 0.43        | 0.59        | 97         | 42        | –      | <b>0</b> |
| Model 3 hybrid 50% beam                     | 0.74        | 73        | 25        | 7.0        | 0.41        | 0.56        | 97         | 45        | –      | <b>0</b> |
| Model 3 hybrid 100% beam                    | 0.74        | 73        | 24        | 7.0        | 0.41        | 0.56        | 97         | 45        | –      | <b>0</b> |

**Table 14:** Test, written items (medicine leaflets).

| System                                      | P(same)     | Kept%     | Joint%    | Clause↓     | Level↓      | Hard w.↓    | Num.%      | Easy=% | Prot.% | Neg.Δ%   |
|---------------------------------------------|-------------|-----------|-----------|-------------|-------------|-------------|------------|--------|--------|----------|
| Copy (no change)                            | 1.00        | 100       | 0         | 0.0         | 0.00        | 0.00        | 100        | –      | –      | 0        |
| App today: AraT5 (int8)                     | 0.83        | 83        | 4         | -1.0        | 0.00        | <b>0.62</b> | <b>100</b> | –      | –      | 13       |
| App today: AraBART (int8)                   | 0.70        | 70        | 9         | 0.0         | 0.00        | 0.12        | <b>100</b> | –      | –      | 14       |
| Model 1                                     | 0.93        | 91        | 0         | 0.0         | 0.00        | 0.12        | <b>100</b> | –      | –      | <b>0</b> |
| Model 2 AraT5 [S2]                          | 0.69        | 65        | 4         | 2.0         | 0.09        | <b>0.62</b> | <b>100</b> | –      | –      | 13       |
| Model 2 AraT5 no tag                        | 0.75        | 74        | 4         | 2.0         | 0.04        | <b>0.62</b> | <b>100</b> | –      | –      | 13       |
| Model 2 AraBART [S2]                        | 0.52        | 52        | 4         | 4.0         | 0.13        | 0.12        | <b>100</b> | –      | –      | 20       |
| Model 2 AraBART no tag                      | 0.61        | 61        | 9         | 3.0         | 0.04        | -0.12       | <b>100</b> | –      | –      | 14       |
| AraBART on corpus v1                        | 0.43        | 43        | 4         | 5.0         | <b>0.43</b> | <b>0.62</b> | <b>100</b> | –      | –      | 27       |
| Model 3 greedy                              | 0.72        | 74        | <b>17</b> | 7.0         | 0.35        | 0.38        | <b>100</b> | –      | –      | 31       |
| Model 3 greedy + net                        | 0.89        | 91        | 13        | 7.0         | 0.22        | 0.00        | <b>100</b> | –      | –      | <b>0</b> |
| Model 3 beam 4                              | 0.82        | 83        | 13        | 0.0         | 0.17        | 0.38        | <b>100</b> | –      | –      | 27       |
| Model 3 beam 4 + net                        | 0.91        | 91        | 9         | 0.0         | 0.04        | 0.00        | <b>100</b> | –      | –      | <b>0</b> |
| Model 3 + DPO                               | 0.76        | 78        | <b>17</b> | <b>10.0</b> | 0.17        | 0.50        | <b>100</b> | –      | –      | 38       |
| Model 3 + DPO + net                         | 0.91        | 91        | 9         | 0.0         | 0.09        | 0.12        | <b>100</b> | –      | –      | <b>0</b> |
| Model 3 int8 per-tensor                     | 0.74        | 74        | 9         | 7.0         | 0.22        | 0.50        | <b>100</b> | –      | –      | 20       |
| Model 3 int8 per-tensor + net               | 0.83        | 83        | 9         | 7.0         | 0.22        | 0.00        | <b>100</b> | –      | –      | <b>0</b> |
| Model 3 app int8 greedy                     | 0.86        | 87        | 9         | 0.0         | 0.09        | 0.50        | <b>100</b> | –      | –      | 13       |
| Model 3 app int8 greedy + net               | 0.91        | 91        | 4         | 0.0         | 0.00        | 0.00        | <b>100</b> | –      | –      | <b>0</b> |
| Model 3 app int8 beam 4                     | 0.87        | 87        | 4         | 0.0         | 0.17        | 0.25        | <b>100</b> | –      | –      | 7        |
| <b>Model 3 app int8 beam 4 + net (ship)</b> | 0.91        | 91        | 4         | 0.0         | 0.09        | 0.00        | <b>100</b> | –      | –      | <b>0</b> |
| Retrain greedy                              | 0.73        | 74        | 13        | 7.0         | 0.30        | <b>0.62</b> | <b>100</b> | –      | –      | 33       |
| Retrain greedy + net                        | 0.87        | 87        | 9         | 0.0         | 0.22        | 0.25        | <b>100</b> | –      | –      | <b>0</b> |
| Retrain beam 4                              | 0.82        | 83        | 13        | 0.0         | 0.26        | <b>0.62</b> | <b>100</b> | –      | –      | 27       |
| Retrain beam 4 + net                        | <b>0.96</b> | <b>96</b> | 9         | 0.0         | 0.13        | 0.25        | <b>100</b> | –      | –      | <b>0</b> |
| Retrain app int8 greedy                     | 0.67        | 70        | 9         | 0.0         | 0.26        | <b>0.62</b> | <b>100</b> | –      | –      | 20       |
| Retrain app int8 greedy + net               | 0.82        | 83        | 4         | 0.0         | 0.13        | 0.12        | <b>100</b> | –      | –      | <b>0</b> |
| Retrain app int8 beam 4                     | 0.82        | 87        | 13        | 0.0         | 0.17        | <b>0.62</b> | <b>100</b> | –      | –      | 20       |
| Retrain app int8 beam 4 + net               | 0.84        | 87        | 9         | 0.0         | 0.13        | 0.12        | <b>100</b> | –      | –      | <b>0</b> |
| Model 3 hybrid 0% beam                      | 0.89        | 91        | 13        | 7.0         | 0.22        | 0.00        | <b>100</b> | –      | –      | <b>0</b> |
| Model 3 hybrid 10% beam                     | 0.89        | 91        | 13        | 7.0         | 0.22        | 0.00        | <b>100</b> | –      | –      | <b>0</b> |
| Model 3 hybrid 20% beam                     | 0.89        | 91        | 13        | 7.0         | 0.22        | 0.00        | <b>100</b> | –      | –      | <b>0</b> |
| Model 3 hybrid 30% beam                     | 0.89        | 91        | 13        | 7.0         | 0.22        | 0.00        | <b>100</b> | –      | –      | <b>0</b> |
| Model 3 hybrid 50% beam                     | 0.90        | 91        | 13        | 7.0         | 0.13        | 0.00        | <b>100</b> | –      | –      | <b>0</b> |
| Model 3 hybrid 100% beam                    | 0.91        | 91        | 9         | 0.0         | 0.04        | 0.00        | <b>100</b> | –      | –      | <b>0</b> |

**Table 15:** Dev, held-out items.

| System                                      | P(same)     | Kept%     | Joint%    | Clause↓     | Level↓      | Hard w.↓    | Num.%      | Easy=%    | Prot.% | Neg.Δ%   |
|---------------------------------------------|-------------|-----------|-----------|-------------|-------------|-------------|------------|-----------|--------|----------|
| Copy (no change)                            | 1.00        | 100       | 0         | 0.0         | 0.00        | 0.00        | 100        | 100       | –      | 0        |
| App today: AraT5 (int8)                     | <b>0.86</b> | <b>86</b> | 4         | 2.4         | 0.08        | 0.33        | <b>100</b> | 31        | –      | 5        |
| App today: AraBART (int8)                   | 0.67        | 66        | 9         | 3.1         | 0.15        | 0.37        | <b>100</b> | 24        | –      | 17       |
| Model 1                                     | 0.82        | 82        | 0         | 0.2         | 0.09        | 0.23        | <b>100</b> | <b>84</b> | –      | <b>0</b> |
| Model 2 AraT5 [S2]                          | 0.69        | 69        | 16        | 6.2         | 0.26        | 0.54        | <b>100</b> | 9         | –      | 14       |
| Model 2 AraT5 no tag                        | 0.81        | 82        | 14        | 3.8         | 0.22        | 0.34        | <b>100</b> | 26        | –      | 9        |
| Model 2 AraBART [S2]                        | 0.54        | 54        | 12        | 6.0         | 0.28        | 0.54        | <b>100</b> | 10        | –      | 21       |
| Model 2 AraBART no tag                      | 0.65        | 65        | 11        | 4.1         | 0.25        | 0.47        | <b>100</b> | 24        | –      | 17       |
| AraBART on corpus v1                        | 0.65        | 65        | 12        | 11.1        | 0.47        | 0.68        | 85         | 38        | –      | 32       |
| Model 3 greedy                              | 0.71        | 71        | 23        | 12.2        | 0.43        | 0.62        | 90         | 29        | –      | 30       |
| Model 3 greedy + net                        | 0.76        | 77        | 21        | 10.6        | 0.36        | 0.56        | 95         | 33        | –      | <b>0</b> |
| Model 3 beam 4                              | 0.73        | 74        | 23        | 11.6        | 0.36        | 0.58        | 95         | 33        | –      | 30       |
| Model 3 beam 4 + net                        | 0.78        | 78        | 20        | 9.7         | 0.27        | 0.53        | <b>100</b> | 36        | –      | <b>0</b> |
| Model 3 + DPO                               | 0.70        | 69        | 25        | 12.3        | 0.34        | 0.61        | 85         | 29        | –      | 33       |
| Model 3 + DPO + net                         | 0.74        | 73        | 22        | 11.8        | 0.25        | 0.59        | 85         | 33        | –      | <b>0</b> |
| Model 3 int8 per-tensor                     | 0.66        | 65        | 8         | 11.2        | 0.36        | 0.56        | 85         | 53        | –      | 29       |
| Model 3 int8 per-tensor + net               | 0.70        | 69        | 7         | 9.3         | 0.36        | 0.51        | 95         | 57        | –      | <b>0</b> |
| Model 3 app int8 greedy                     | 0.71        | 72        | 14        | 11.3        | 0.30        | 0.57        | 95         | 52        | –      | 35       |
| Model 3 app int8 greedy + net               | 0.75        | 75        | 10        | 10.4        | 0.14        | 0.56        | 95         | 53        | –      | <b>0</b> |
| Model 3 app int8 beam 4                     | 0.77        | 76        | 17        | 11.2        | 0.36        | 0.48        | 90         | 60        | –      | 25       |
| <b>Model 3 app int8 beam 4 + net (ship)</b> | 0.80        | 79        | 14        | 10.1        | 0.26        | 0.46        | 90         | 62        | –      | <b>0</b> |
| Retrain greedy                              | 0.70        | 70        | 25        | <b>12.7</b> | <b>0.51</b> | <b>0.72</b> | 90         | 34        | –      | 38       |
| Retrain greedy + net                        | 0.75        | 75        | 22        | 10.1        | 0.40        | 0.68        | 95         | 36        | –      | <b>0</b> |
| Retrain beam 4                              | 0.77        | 77        | <b>27</b> | 12.4        | 0.42        | 0.67        | <b>100</b> | 40        | –      | 30       |
| Retrain beam 4 + net                        | 0.80        | 80        | 24        | 10.8        | 0.34        | 0.64        | <b>100</b> | 41        | –      | <b>0</b> |
| Retrain app int8 greedy                     | 0.76        | 76        | 18        | 10.6        | 0.39        | 0.64        | 85         | 57        | –      | 26       |
| Retrain app int8 greedy + net               | 0.79        | 79        | 15        | 8.3         | 0.33        | 0.60        | <b>100</b> | 57        | –      | <b>0</b> |
| Retrain app int8 beam 4                     | 0.80        | 80        | 18        | 9.6         | 0.29        | 0.58        | 90         | 60        | –      | 18       |
| Retrain app int8 beam 4 + net               | 0.82        | 83        | 15        | 8.0         | 0.23        | 0.48        | 95         | 60        | –      | <b>0</b> |
| Model 3 hybrid 0% beam                      | 0.76        | 77        | 21        | 10.6        | 0.36        | 0.56        | 95         | 33        | –      | <b>0</b> |
| Model 3 hybrid 10% beam                     | 0.75        | 76        | 24        | 10.3        | 0.32        | 0.54        | 95         | 33        | –      | <b>0</b> |
| Model 3 hybrid 20% beam                     | 0.76        | 77        | 22        | 10.3        | 0.30        | 0.55        | 95         | 31        | –      | <b>0</b> |
| Model 3 hybrid 30% beam                     | 0.77        | 77        | 23        | 10.1        | 0.30        | 0.56        | <b>100</b> | 31        | –      | <b>0</b> |
| Model 3 hybrid 50% beam                     | 0.77        | 78        | 20        | 9.7         | 0.27        | 0.54        | <b>100</b> | 36        | –      | <b>0</b> |
| Model 3 hybrid 100% beam                    | 0.78        | 78        | 20        | 9.7         | 0.27        | 0.53        | <b>100</b> | 36        | –      | <b>0</b> |

Table 16: Dev, outside items.

| System                                      | P(same)     | Kept%     | Joint%    | Clause↓     | Level↓      | Hard w.↓    | Num.%      | Easy=%    | Prot.% | Neg.Δ%   |
|---------------------------------------------|-------------|-----------|-----------|-------------|-------------|-------------|------------|-----------|--------|----------|
| Copy (no change)                            | 1.00        | 100       | 0         | 0.0         | 0.00        | 0.00        | 100        | 100       | –      | 0        |
| App today: AraT5 (int8)                     | 0.74        | <b>74</b> | 7         | 2.1         | 0.09        | 0.29        | <b>100</b> | 46        | –      | 12       |
| App today: AraBART (int8)                   | 0.52        | 52        | 14        | 4.8         | 0.29        | 0.48        | <b>100</b> | 25        | –      | 27       |
| Model 1                                     | 0.72        | 72        | 1         | 0.3         | 0.05        | 0.14        | 92         | 62        | –      | <b>0</b> |
| Model 2 AraT5 [S2]                          | 0.57        | 58        | 15        | 3.6         | 0.39        | 0.56        | 85         | 17        | –      | 25       |
| Model 2 AraT5 no tag                        | 0.69        | 69        | 18        | 3.2         | 0.31        | 0.44        | 92         | 33        | –      | 22       |
| Model 2 AraBART [S2]                        | 0.46        | 47        | 14        | 6.0         | 0.29        | 0.58        | 85         | 25        | –      | 29       |
| Model 2 AraBART no tag                      | 0.53        | 52        | 10        | 4.3         | 0.32        | 0.48        | <b>100</b> | 25        | –      | 27       |
| AraBART on corpus v1                        | 0.58        | 58        | 11        | 8.3         | <b>0.77</b> | 0.59        | 92         | 54        | –      | 43       |
| Model 3 greedy                              | 0.57        | 57        | 17        | 9.2         | 0.75        | <b>0.82</b> | 92         | 33        | –      | 31       |
| Model 3 greedy + net                        | 0.69        | 69        | 13        | 6.8         | 0.49        | 0.59        | 92         | 38        | –      | <b>0</b> |
| Model 3 beam 4                              | 0.60        | 59        | 13        | 9.4         | 0.47        | 0.77        | <b>100</b> | 38        | –      | 38       |
| Model 3 beam 4 + net                        | 0.70        | 69        | 12        | 7.0         | 0.28        | 0.63        | <b>100</b> | 42        | –      | <b>0</b> |
| Model 3 + DPO                               | 0.53        | 54        | <b>19</b> | <b>10.1</b> | 0.40        | 0.68        | 92         | 29        | –      | 40       |
| Model 3 + DPO + net                         | 0.63        | 63        | 15        | 7.9         | 0.27        | 0.54        | 92         | 33        | –      | <b>0</b> |
| Model 3 int8 per-tensor                     | 0.58        | 59        | 12        | 8.3         | 0.53        | 0.60        | 92         | 54        | –      | 32       |
| Model 3 int8 per-tensor + net               | 0.66        | 67        | 9         | 5.9         | 0.41        | 0.47        | <b>100</b> | 62        | –      | <b>0</b> |
| Model 3 app int8 greedy                     | 0.59        | 61        | 14        | 7.5         | 0.61        | 0.51        | 85         | 50        | –      | 28       |
| Model 3 app int8 greedy + net               | 0.68        | 70        | 12        | 5.9         | 0.46        | 0.38        | <b>100</b> | 50        | –      | <b>0</b> |
| Model 3 app int8 beam 4                     | 0.68        | 67        | 11        | 7.3         | 0.37        | 0.44        | 92         | <b>71</b> | –      | 27       |
| <b>Model 3 app int8 beam 4 + net (ship)</b> | <b>0.75</b> | <b>74</b> | 9         | 5.9         | 0.31        | 0.36        | <b>100</b> | <b>71</b> | –      | <b>0</b> |
| Retrain greedy                              | 0.60        | 59        | 15        | 8.7         | 0.68        | 0.73        | 92         | 54        | –      | 24       |
| Retrain greedy + net                        | 0.69        | 69        | 11        | 7.3         | 0.56        | 0.58        | <b>100</b> | 62        | –      | <b>0</b> |
| Retrain beam 4                              | 0.65        | 64        | 16        | 9.0         | 0.45        | 0.74        | <b>100</b> | 54        | –      | 26       |
| Retrain beam 4 + net                        | 0.70        | 69        | 14        | 6.9         | 0.42        | 0.65        | <b>100</b> | 58        | –      | <b>0</b> |
| Retrain app int8 greedy                     | 0.60        | 60        | 14        | 8.8         | 0.53        | 0.62        | 92         | 58        | –      | 34       |
| Retrain app int8 greedy + net               | 0.68        | 68        | 10        | 6.2         | 0.44        | 0.52        | <b>100</b> | 62        | –      | <b>0</b> |
| Retrain app int8 beam 4                     | 0.66        | 66        | 16        | 7.3         | 0.41        | 0.60        | <b>100</b> | <b>71</b> | –      | 30       |
| Retrain app int8 beam 4 + net               | 0.70        | 70        | 12        | 5.2         | 0.33        | 0.53        | <b>100</b> | <b>71</b> | –      | <b>0</b> |
| Model 3 hybrid 0% beam                      | 0.69        | 69        | 13        | 6.8         | 0.49        | 0.59        | 92         | 38        | –      | <b>0</b> |
| Model 3 hybrid 10% beam                     | 0.69        | 69        | 14        | 6.8         | 0.46        | 0.62        | 92         | 33        | –      | <b>0</b> |
| Model 3 hybrid 20% beam                     | 0.70        | 70        | 14        | 7.0         | 0.45        | 0.58        | 92         | 33        | –      | <b>0</b> |
| Model 3 hybrid 30% beam                     | 0.71        | 71        | 14        | 7.0         | 0.41        | 0.63        | 92         | 38        | –      | <b>0</b> |
| Model 3 hybrid 50% beam                     | 0.69        | 69        | 14        | 7.3         | 0.33        | 0.63        | <b>100</b> | 38        | –      | <b>0</b> |
| Model 3 hybrid 100% beam                    | 0.70        | 69        | 12        | 7.0         | 0.28        | 0.63        | <b>100</b> | 42        | –      | <b>0</b> |

Table 17: Dev, written items.

| System                                      | P(same)     | Kept%      | Joint% | Clause↓ | Level↓ | Hard w.↓ | Num.% | Easy=% | Prot.% | Neg.Δ% |
|---------------------------------------------|-------------|------------|--------|---------|--------|----------|-------|--------|--------|--------|
| Copy (no change)                            | 1.00        | 100        | 0      | 0.0     | 0.00   | 0.00     | 100   | –      | –      | 0      |
| App today: AraT5 (int8)                     | 0.86        | 85         | 7      | 1.7     | 0.00   | 0.36     | 100   | –      | –      | 12     |
| App today: AraBART (int8)                   | 0.56        | 52         | 4      | 3.3     | 0.11   | 0.18     | 100   | –      | –      | 18     |
| Model 1                                     | 0.96        | 96         | 0      | 0.0     | 0.00   | 0.00     | 100   | –      | –      | 0      |
| Model 2 AraT5 [S2]                          | 0.69        | 63         | 11     | 4.0     | 0.11   | 0.36     | 100   | –      | –      | 6      |
| Model 2 AraT5 no tag                        | 0.73        | 67         | 7      | 2.0     | 0.00   | 0.45     | 100   | –      | –      | 12     |
| Model 2 AraBART [S2]                        | 0.40        | 37         | 0      | 2.7     | 0.15   | 0.18     | 67    | –      | –      | 24     |
| Model 2 AraBART no tag                      | 0.49        | 48         | 4      | 3.3     | 0.07   | 0.18     | 100   | –      | –      | 24     |
| AraBART on corpus v1                        | 0.70        | 67         | 7      | 3.3     | 0.37   | 0.27     | 100   | –      | –      | 12     |
| Model 3 greedy                              | 0.70        | 63         | 30     | 4.0     | 0.41   | 0.36     | 100   | –      | –      | 24     |
| Model 3 greedy + net                        | 0.84        | 81         | 26     | 3.7     | 0.19   | 0.27     | 100   | –      | –      | 0      |
| Model 3 beam 4                              | 0.84        | 78         | 19     | 4.0     | 0.26   | 0.18     | 100   | –      | –      | 6      |
| Model 3 beam 4 + net                        | 0.95        | 89         | 19     | 4.0     | 0.11   | 0.18     | 100   | –      | –      | 0      |
| Model 3 + DPO                               | 0.76        | 74         | 22     | 6.0     | 0.26   | 0.27     | 100   | –      | –      | 24     |
| Model 3 + DPO + net                         | 0.91        | 89         | 15     | 3.7     | 0.07   | 0.18     | 100   | –      | –      | 0      |
| Model 3 int8 per-tensor                     | 0.70        | 70         | 22     | 7.0     | 0.33   | 0.27     | 100   | –      | –      | 12     |
| Model 3 int8 per-tensor + net               | 0.85        | 85         | 19     | 4.7     | 0.11   | 0.27     | 100   | –      | –      | 0      |
| Model 3 app int8 greedy                     | 0.81        | 81         | 15     | 3.7     | 0.26   | 0.27     | 100   | –      | –      | 6      |
| Model 3 app int8 greedy + net               | 0.89        | 89         | 15     | 3.7     | 0.11   | 0.27     | 100   | –      | –      | 0      |
| Model 3 app int8 beam 4                     | 0.92        | 93         | 19     | 3.7     | 0.22   | 0.27     | 100   | –      | –      | 6      |
| <b>Model 3 app int8 beam 4 + net (ship)</b> | <b>1.00</b> | <b>100</b> | 15     | 3.7     | 0.04   | 0.27     | 100   | –      | –      | 0      |
| Retrain greedy                              | 0.49        | 44         | 26     | 6.0     | 0.44   | 0.45     | 67    | –      | –      | 12     |
| Retrain greedy + net                        | 0.64        | 59         | 15     | 6.0     | 0.22   | 0.18     | 100   | –      | –      | 0      |
| Retrain beam 4                              | 0.79        | 70         | 30     | 4.3     | 0.48   | 0.45     | 100   | –      | –      | 6      |
| Retrain beam 4 + net                        | 0.86        | 78         | 19     | 4.3     | 0.26   | 0.36     | 100   | –      | –      | 0      |
| Retrain app int8 greedy                     | 0.67        | 63         | 30     | 2.3     | 0.52   | 0.55     | 100   | –      | –      | 24     |
| Retrain app int8 greedy + net               | 0.81        | 78         | 15     | 2.3     | 0.19   | 0.36     | 100   | –      | –      | 0      |
| Retrain app int8 beam 4                     | 0.79        | 74         | 11     | 2.3     | 0.22   | 0.36     | 100   | –      | –      | 12     |
| Retrain app int8 beam 4 + net               | 0.89        | 85         | 11     | 2.3     | 0.22   | 0.36     | 100   | –      | –      | 0      |
| Model 3 hybrid 0% beam                      | 0.84        | 81         | 26     | 3.7     | 0.19   | 0.27     | 100   | –      | –      | 0      |
| Model 3 hybrid 10% beam                     | 0.84        | 81         | 26     | 3.7     | 0.19   | 0.27     | 100   | –      | –      | 0      |
| Model 3 hybrid 20% beam                     | 0.84        | 81         | 26     | 3.7     | 0.19   | 0.27     | 100   | –      | –      | 0      |
| Model 3 hybrid 30% beam                     | 0.84        | 81         | 26     | 3.7     | 0.19   | 0.27     | 100   | –      | –      | 0      |
| Model 3 hybrid 50% beam                     | 0.85        | 81         | 22     | 3.7     | 0.15   | 0.27     | 100   | –      | –      | 0      |
| Model 3 hybrid 100% beam                    | 0.95        | 89         | 19     | 4.0     | 0.11   | 0.18     | 100   | –      | –      | 0      |

## B Paired differences against today’s app, every system

Test, core items: system minus today’s app (paired; ↑/↓: interval excludes zero)

| System – app                         | P(same) | Kept (pts) | Clause | Level  | Numbers (pts) |
|--------------------------------------|---------|------------|--------|--------|---------------|
| Copy (no change)                     | +0.21↑  | +21↑       | -1.9↓  | -0.14↓ | +16↑          |
| App today: AraBART (int8)            | -0.28↓  | -28↓       | +2.8↑  | +0.15↑ | -8            |
| Model 1                              | -0.03   | -3         | -1.9↓  | -0.05  | +3            |
| Model 2 AraT5 [S2]                   | -0.19↓  | -19↓       | +4.4↑  | +0.26↑ | -1            |
| Model 2 AraT5 no tag                 | -0.10↓  | -10↓       | +2.6↑  | +0.04  | -5            |
| Model 2 AraBART [S2]                 | -0.38↓  | -39↓       | +4.8↑  | +0.24↑ | -5            |
| Model 2 AraBART no tag               | -0.27↓  | -27↓       | +3.0↑  | +0.12↑ | -8            |
| AraBART on corpus v1                 | -0.16↓  | -16↓       | +6.2↑  | +0.29↑ | +4            |
| Model 3 greedy                       | -0.10↓  | -9↓        | +7.6↑  | +0.35↑ | +9↑           |
| Model 3 greedy + net                 | -0.04   | -3         | +5.7↑  | +0.24↑ | +16↑          |
| Model 3 beam 4                       | -0.08↓  | -7↓        | +7.1↑  | +0.34↑ | +8↑           |
| Model 3 beam 4 + net                 | -0.01   | -0         | +4.9↑  | +0.23↑ | +16↑          |
| Model 3 + DPO                        | -0.13↓  | -13↓       | +7.4↑  | +0.22↑ | +3            |
| Model 3 + DPO + net                  | -0.04   | -4         | +5.4↑  | +0.13↑ | +12↑          |
| Model 3 int8 per-tensor              | -0.11↓  | -11↓       | +6.7↑  | +0.21↑ | +4            |
| Model 3 int8 per-tensor + net        | -0.02   | -2         | +4.6↑  | +0.11↑ | +16↑          |
| Model 3 app int8 greedy              | -0.08↓  | -8↓        | +6.1↑  | +0.20↑ | +5            |
| Model 3 app int8 greedy + net        | +0.00   | +0         | +4.3↑  | +0.12↑ | +16↑          |
| Model 3 app int8 beam 4              | -0.03   | -3         | +5.2↑  | +0.10↑ | +7            |
| Model 3 app int8 beam 4 + net (ship) | +0.04   | +4         | +3.5↑  | +0.07  | +16↑          |
| Retrain greedy                       | -0.09↓  | -8↓        | +6.9↑  | +0.34↑ | +8↑           |
| Retrain greedy + net                 | -0.02   | -1         | +4.7↑  | +0.23↑ | +16↑          |
| Retrain beam 4                       | -0.06↓  | -6↓        | +5.9↑  | +0.27↑ | +7            |

| System – app                  | P(same) | Kept (pts) | Clause | Level  | Numbers (pts) |
|-------------------------------|---------|------------|--------|--------|---------------|
| Retrain beam 4 + net          | +0.00   | +0         | +3.9↑  | +0.19↑ | +16↑          |
| Retrain app int8 greedy       | -0.03   | -3         | +4.3↑  | +0.11↑ | +4            |
| Retrain app int8 greedy + net | +0.04↑  | +4         | +2.6↑  | +0.07  | +16↑          |
| Retrain app int8 beam 4       | +0.01   | +1         | +3.6↑  | +0.03  | +5            |
| Retrain app int8 beam 4 + net | +0.06↑  | +6↑        | +2.2↑  | +0.00  | +13↑          |
| Model 3 hybrid 0% beam        | -0.04   | -3         | +5.7↑  | +0.24↑ | +16↑          |
| Model 3 hybrid 10% beam       | -0.04↓  | -3         | +5.8↑  | +0.24↑ | +16↑          |
| Model 3 hybrid 20% beam       | -0.03   | -2         | +5.7↑  | +0.24↑ | +16↑          |
| Model 3 hybrid 30% beam       | -0.02   | -2         | +5.6↑  | +0.24↑ | +16↑          |
| Model 3 hybrid 50% beam       | -0.01   | -1         | +5.2↑  | +0.23↑ | +16↑          |
| Model 3 hybrid 100% beam      | -0.01   | -0         | +4.9↑  | +0.23↑ | +16↑          |

## Dev, core items

| System – app                         | P(same) | Kept (pts) | Clause | Level  | Numbers (pts) |
|--------------------------------------|---------|------------|--------|--------|---------------|
| Copy (no change)                     | +0.21↑  | +22↑       | -1.4↓  | -0.01  | +28↑          |
| App today: AraBART (int8)            | -0.24↓  | -24↓       | +2.4↑  | +0.20↑ | +0            |
| Model 1                              | -0.01   | -1         | -0.9   | -0.03  | +12           |
| Model 2 AraT5 [S2]                   | -0.16↓  | -16↓       | +5.0↑  | +0.18↑ | +4            |
| Model 2 AraT5 no tag                 | -0.01   | -1         | +2.3↑  | -0.01  | +0            |
| Model 2 AraBART [S2]                 | -0.37↓  | -37↓       | +4.6↑  | +0.24↑ | +0            |
| Model 2 AraBART no tag               | -0.23↓  | -23↓       | +2.0↑  | +0.15↑ | +4            |
| AraBART on corpus v1                 | -0.16↓  | -16↓       | +6.5↑  | +0.47↑ | +12           |
| Model 3 greedy                       | -0.06   | -6         | +7.3↑  | +0.45↑ | +8            |
| Model 3 greedy + net                 | -0.01   | -1         | +5.3↑  | +0.32↑ | +28↑          |
| Model 3 beam 4                       | -0.03   | -2         | +7.4↑  | +0.44↑ | +12           |
| Model 3 beam 4 + net                 | +0.01   | +2         | +5.5↑  | +0.35↑ | +28↑          |
| Model 3 + DPO                        | -0.08↓  | -8↓        | +7.2↑  | +0.33↑ | +12           |
| Model 3 + DPO + net                  | -0.03   | -3         | +5.0↑  | +0.29↑ | +28↑          |
| Model 3 int8 per-tensor              | -0.07   | -6         | +6.5↑  | +0.31↑ | +12           |
| Model 3 int8 per-tensor + net        | -0.03   | -2         | +5.4↑  | +0.25↑ | +24↑          |
| Model 3 app int8 greedy              | -0.05   | -4         | +7.0↑  | +0.30↑ | +16           |
| Model 3 app int8 greedy + net        | +0.00   | +1         | +4.9↑  | +0.24↑ | +28↑          |
| Model 3 app int8 beam 4              | +0.03   | +3         | +5.7↑  | +0.19↑ | +20↑          |
| Model 3 app int8 beam 4 + net (ship) | +0.06   | +7         | +4.0↑  | +0.15↑ | +28↑          |
| Retrain greedy                       | -0.06   | -4         | +5.9↑  | +0.38↑ | +16           |
| Retrain greedy + net                 | -0.00   | +1         | +3.8↑  | +0.29↑ | +24↑          |
| Retrain beam 4                       | -0.04   | -3         | +5.9↑  | +0.36↑ | +20           |
| Retrain beam 4 + net                 | +0.01   | +2         | +3.8↑  | +0.29↑ | +28↑          |
| Retrain app int8 greedy              | -0.02   | -2         | +4.6↑  | +0.24↑ | +8            |
| Retrain app int8 greedy + net        | +0.01   | +2         | +3.4↑  | +0.20↑ | +20           |
| Retrain app int8 beam 4              | +0.01   | +1         | +3.3↑  | +0.24↑ | +20↑          |
| Retrain app int8 beam 4 + net        | +0.04   | +4         | +2.6↑  | +0.20↑ | +28↑          |
| Model 3 hybrid 0% beam               | -0.01   | -1         | +5.3↑  | +0.32↑ | +28↑          |
| Model 3 hybrid 10% beam              | -0.02   | -2         | +5.9↑  | +0.34↑ | +28↑          |
| Model 3 hybrid 20% beam              | -0.03   | -2         | +6.1↑  | +0.35↑ | +28↑          |
| Model 3 hybrid 30% beam              | -0.02   | -2         | +6.1↑  | +0.38↑ | +28↑          |
| Model 3 hybrid 50% beam              | +0.00   | +1         | +5.7↑  | +0.36↑ | +28↑          |
| Model 3 hybrid 100% beam             | +0.01   | +2         | +5.5↑  | +0.35↑ | +28↑          |

## C Reproducing this report

docs/benchmark\_report/README.md explains how to rebuild every system’s outputs, the Gemma 31B scores and the scorecards from scratch. With the outputs and score files in place:

```
export BAYANBENCH_DATA=<snapshot of Congi-libya/bayanbench-data> # v1.0 data
export BAYAN_DATA=<repo>/data/processed/meaning_reward # system outputs and score files
export SAMER_DIR=<licensed SAMER copy> # hard-word measures (optional)
```

```
python collect.py    # official v2.0 scorecards, every system and split
make                 # analyse.py -> figures.py -> tables.py -> latexmk
```